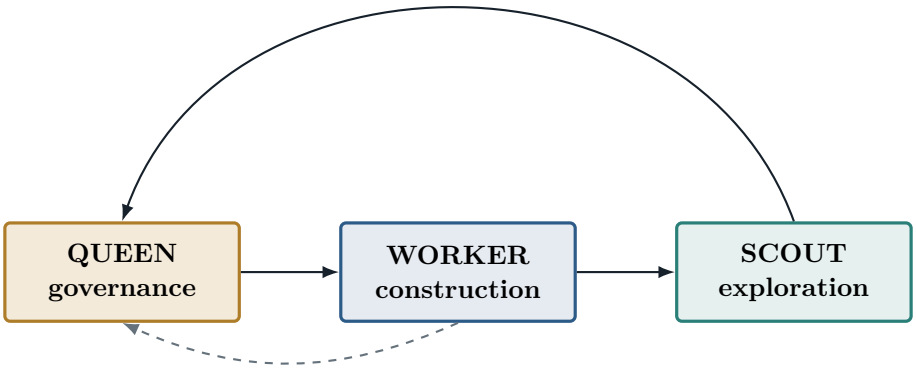
Ant Hill Reasoning (AHR)

Unified Mathematical Theory

Hierarchical energy-based reasoning with Queen, Worker, Scout, stigmergic memory, hard verification, and principled abstention



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This monograph consolidates the AHR mathematical foundation, original recurrent/stigmergic design, verifier-backed CSP and proof-search theory, and the finite formal model.

Abstract

Ant Hill Reasoning (AHR) is formalized here as a verifier-constrained, energy-based search over structured reasoning trajectories whose state and action spaces contain three primitive roles: a slow Queen governing strategy, budget, pruning, verification, and stopping; a Worker constructing local derivations or assignments; and a fast Scout proposing branches and gathering evidence. The architecture combines the multi-timescale recurrence of hierarchical reasoning models, the variational and optimization viewpoint of energy-based models, and a bounded stigmergic memory inspired by pheromone dynamics. The defining object is not the biological metaphor but the Queen/Worker/Scout compatibility relation

$$\mathcal{A}_{\mathrm{QWS}} \subseteq Q \times W \times U \times S \times A,$$

which specifies when the three roles are coordinated.

The document gives the complete finite base theory: semantic solution relation, role spaces, dynamic and strict transition relations, legal trajectories, terminal extraction, verifier, hard and soft energy terms, bounded candidate sets, deterministic tie-breaking, abstention, and role-factored search. It proves finite, compact, and finite-sublevel argmin existence; verifier-relative and domain soundness; abstention safety; alignment coherence; energy-margin recovery; perturbation stability; safe Queen pruning; deterministic and stochastic completeness under Scout coverage; bounded pheromone stability; and a conditional hierarchical work advantage. A fully finite modus-ponens model keeps semantic answers independent of proof certificates and proves checker and verifier soundness by induction. The same substrate is then instantiated for finite CSPs, temporal constraints, proof-assistant search, symbolic Diophantine proof search, and Sudoku-style structured prediction. A self-contained reader's toolkit introduces every non-elementary convention needed by an undergraduate in mathematics or computer science.

The central definition is:

AHR is Queen/Worker/Scout hierarchical energy minimization over verified reasoning trajectories.

Status of Claims

This document deliberately separates five kinds of statement.

Status	Meaning
Definition	A mathematical choice that fixes what an AHR system is.
Theorem	A proved implication under explicit assumptions stated in this document.
Instantiation	A domain-specific realization, such as the finite proof model or a finite CSP.
Empirical observation	A measurement from a named saved artifact. It is evidence about that run, not a theorem about AHR in general.
Conjecture	A research claim not established by the base theory, including broad generalization, learned-energy convergence, and universal-reasoner claims.

In particular, a fixed parameter count with respect to recurrent depth is not constant-time inference, verifier-relative soundness is not domain soundness unless the verifier itself is proved sound, and a successful verifier-backed experiment is not a proof of search completeness or out-of-distribution generalization.

Contents

Abstract i

Status of Claims ii

How to Read This Monograph ix

Mathematical and Computer-Science Toolkit xi

I Foundations 1

1 Origin and Design Commitments 2

1.1 The synthesis . . . 2

1.2 Non-negotiable architecture . . . 2

1.3 Design requirements . . . 3

2 Semantic Contract 4

2.1 Problems, answers, and truth . . . 4

2.2 Three correctness levels . . . 5

2.3 Abstention semantics . . . 5

3 The AHR Mathematical Object 6

3.1 Primitive spaces . . . 6

3.2 Dynamics and trajectories . . . 6

3.3 Budget and policy . . . 7

3.4 Compact specification . . . 8

4 Queen, Worker, and Scout as Primitive Structure 9

4.1 Queen state . . . 9

4.2 Worker state . . . 9

4.3 Scout state . . . 10
4.4 Nested time scales . . . 10

5 The QWS Compatibility Relation 12
5.1 Auxiliary maps . . . 12
5.2 Exact compatibility clauses . . . 13
5.3 Hard and soft alignment . . . 14

6 Stigmergic Memory and Energy-Based Dynamics 15
6.1 Pheromone state . . . 15
6.2 Exponential memory . . . 17
6.3 Pheromone as a variational prior . . . 18

7 The AHR Energy Functional 20
7.1 Extended-real formulation . . . 20
7.2 Hard indicators . . . 21
7.3 Hard constraints are not weights . . . 21

8 Objective, Search, Repair, and Halting 23
8.1 Verified bounded candidates . . . 23
8.2 Role-factored search . . . 23
8.3 Verifier-guided repair . . . 24
8.4 Adaptive computation . . . 24

II Guarantees 25

Guarantee Roadmap 26

9 Existence and Well-Posedness 27

10 Soundness and Abstention 30

11 Alignment, Margins, and Robustness 31
11.1 Two margin statements . . . 31
11.2 Perturbation stability . . . 32

12 Safe Pruning and Completeness 34
12.1 Prefix notation . . . 34

12.2 Deterministic coverage . . . 34
12.3 Stochastic coverage . . . 35

13 Complexity, Training Depth, and Compute 37
13.1 Conditional branching advantage . . . 37
13.2 Fixed parameters versus inference cost . . . 38
13.3 Detached one-step gradients . . . 39

III Complete and Domain-Specific Instantiations 42

14 Finite Modus-Ponens AHR 43
14.1 Independent semantics . . . 43
14.2 Finite certificates and checker . . . 44
14.3 Exact finite Queen, Worker, and Scout spaces . . . 45
14.4 Exact compatibility and dynamics . . . 47
14.5 Extraction, verification, and domain soundness . . . 48
14.6 Valid and invalid trajectories . . . 49

15 Finite Constraint Satisfaction and Temporal Reasoning 51
15.1 CSP substrate . . . 51
15.2 Role mapping . . . 52
15.3 Temporal and revised constraints . . . 52

16 Proof-Assistant Search 53

17 Symbolic and Diophantine Proof Search 55

18 Sudoku and Structured Prediction 57

19 Empirical Evidence and Its Exact Scope 58
19.1 Evidence-audit rule . . . 58
19.2 Sudoku evidence . . . 58
19.2.1 Early archive . . . 58
19.2.2 AHR-lite hybrid notebook . . . 59
19.2.3 Pure-neural Sudoku-extreme run . . . 60
19.3 Triple-role AHR-CSP evidence . . . 60
19.4 Abstaining multi-family ranker . . . 61

19.5 Lean proof-search evidence, including the negative result . . . 61
19.6 What the evidence establishes . . . 62

IV Learning Theory and Research Program 63

20 Learning Hierarchical Energies 64
20.1 Parameterized energy . . . 64
20.2 Ranking consistency target . . . 64
20.3 Calibration and abstention . . . 65

21 Open Mathematical Problems 66

22 Falsifiable AHR Research Program 67
22.1 What the theory already establishes . . . 67
22.2 What must be tested . . . 67
22.3 Minimal ablation lattice . . . 67

23 The Unified Definition 69

A Notation 70

B Hard and Soft Energy Catalogue 71

C Dependency Map of the Theory 72

D Source Synthesis Map 73

List of Figures

1.1 The AHR contract. Search is internal; final correctness is controlled by an external verifier. 3

4.1 Three nested recurrence scales. The count NKT is computational depth, not parameter depth. 11

7.1 The extended-real EBM view. Hard constraints define feasibility; soft energy ranks only feasible candidates. 22

C.1 Logical dependencies of the AHR theory. 72

List of Tables

2 Suggested reading paths. ix

8.1 Logical scope of the principal guarantees. 26

15.1 AHR roles in a finite CSP. 52

A.1 Canonical AHR notation. 70

B.1 Energy semantics. 71

D.1 How the AHR paper corpus contributes to the unified theory. 73

How to Read This Monograph

Audience and prerequisites

The core theory is written for a reader who has completed an introductory course in discrete mathematics and at least one course in programming, algorithms, or mathematical proof. Calculus and linear algebra are needed only for the differentiable-learning sections. Elementary probability is needed for the stochastic-completeness theorem. The compact-space existence theorem uses a small amount of real analysis or topology, but every topological term used there is defined in the next chapter.

Table 2: Suggested reading paths.

Goal	Read first	Then read
Understand the formal AHR contract	Reader's toolkit; Origin and Design Commitments	Semantic Contract through Objective, Search, Repair, and Halting
Check the guarantees	Reader's toolkit; Semantic Contract; The AHR Mathematical Object	All chapters in Part II, in order
Understand the neural/EBM interpretation	Reader's toolkit; Stigmergic Memory; AHR Energy Functional	Complexity and Training; Learning Hierarchical Energies
Understand exact verification	Reader's toolkit; Semantic Contract	Finite Modus-Ponens AHR; CSP; Proof-Assistant Search; Sudoku
Reproduce the logical core only	Reader's toolkit	Semantic Contract; AHR Mathematical Object; Energy Functional; Objective; Soundness; Unified Definition

Four questions to ask at every claim

Every formal statement in this monograph can be audited by asking:

1. **What are the types?** For example, is the object a state, action, trajectory, answer, set, relation, or real number?

2. **What is assumed?** A theorem proves only an implication from its hypotheses. Coverage, sound pruning, and verifier soundness are not automatic properties of an implementation.
3. **What is concluded?** Soundness, completeness, optimality, and low empirical error are different conclusions.
4. **Is the claim formal or empirical?** A theorem follows from definitions and assumptions. A measured rate follows from a particular dataset, split, program, checkpoint, and evaluation procedure.

Proof-reading convention

A displayed implication $P \Rightarrow Q$ should be read as: whenever all conditions collected in P hold, Q must hold. It does not assert that P holds in every AHR implementation. Proofs in this edition expand every algebraic step that carries logical weight. Familiar one-line facts, such as "a finite nonempty set of real numbers has a minimum," are stated explicitly when they are used.

The verifier protects emitted answers only if the verifier is itself sound for the intended domain. Search quality cannot manufacture that missing premise.

Mathematical and Computer-Science Toolkit

Sets, elements, and products

A *set* is a collection of objects. The notation $x \in X$ says that x is an element of the set X. The notation $A \subseteq B$ says that every element of A also belongs to B. These statements have different types: $x \in X$ relates an element to a set, whereas $A \subseteq B$ relates two sets.

The Cartesian product

$$A \times B = \{(a, b) : a \in A,\ b \in B\}$$

is the set of ordered pairs. Thus $(a, b) \in A \times B$ means simultaneously $a \in A$ and $b \in B$. A five-coordinate AHR compatibility tuple has type

$$(q, w, u, s, a) \in Q \times W \times U \times S \times A.$$

If $h = (q, w, u, s) \in H$, then the same information may be written as (h, a), but (h, a) belongs to $H \times A$, not literally to $Q \times W \times U \times S \times A$. Coordinate expansion is required before testing membership in the latter set.

For a set K, the notation 2^K means its power set, the set of all subsets of K. For sets A and B, the notation B^A means the set of all functions from A to B. Therefore

$$F_U \in 2^{\mathcal{K}} \quad \text{and} \quad e_U \in \mathcal{E}_K^{\mathcal{K}}$$

mean, respectively, that F_U is a subset of the branch universe and that e_U assigns one energy-grid value to every branch.

Functions, relations, and correspondences

A function $f : A \to B$ assigns exactly one value $f(a) \in B$ to each $a \in A$. A relation $R \subseteq A \times B$ may associate one input with zero, one, or many outputs. A correspondence $I : X \to 2^H$ is a function whose value $I(x)$ is itself a set; it can therefore specify several admissible initial states.

The verifier has a *dependent domain*:

$$\{(x, \tau, y) : x \in X,\ \tau \in \Omega(x),\ y \in Y\}.$$

The allowed trajectory set depends on the selected problem x. Writing this domain explicitly prevents the verifier from being applied to an ill-typed trajectory from another problem.

Logic and quantifiers

The universal quantifier $\forall$ means "for every," and the existential quantifier $\exists$ means "there exists." The implication $P \Rightarrow Q$ can fail only when P is true and Q is false. The equivalence $P \iff Q$ abbreviates the two implications $P \Rightarrow Q$ and $Q \Rightarrow P$.

The order of quantifiers matters. The coverage statement

$$\forall i \; \exists t : \text{ branch } i \text{ is proposed at time } t$$

does not require one common time for all branches. By contrast, $\exists t \; \forall i$ would.

Finite sequences and trajectories

A trajectory with L transitions is

$$\tau = (h_0, a_0, h_1, \ldots, a_{L-1}, h_L).$$

It contains $L + 1$ states and L actions. This monograph calls L the *transition length*; the symbol $|\tau|$ denotes that value when no ambiguity is possible. The index range $0 \le t < L$ is exactly $t \in \{0, \ldots, L - 1\}$.

Induction on a nonnegative integer proves a claim for all finite stages. It requires a base case, usually stage 0, and a step showing that truth at stage i implies truth at stage $i + 1$. The completeness and checker proofs later in the monograph use this pattern.

Indicators, extended-real energy, and minimization

For a proposition P, the indicator

$$\mathbf{1}\{P\} = \begin{cases} 1, & P \text{ is true,} \\ 0, & P \text{ is false} \end{cases}$$

turns a logical violation into a number. A sum of nonnegative indicators is zero if and only if every summand is zero.

The extended real line used here is

$$\mathbb{R} \cup \{+\infty\}.$$

The value $+\infty$ is not a very large trainable penalty. It means that the candidate is forbidden in the exact optimization problem. For finite $r \in \mathbb{R}$, the arithmetic conventions used here include $r + (+\infty) = +\infty$.

For a function $E : D \to \mathbb{R} \cup \{+\infty\}$,

$$\arg\min_{z \in D} E(z) = \{z \in D : E(z) \le E(z') \text{ for every } z' \in D\}$$

is a set of minimizers. It may contain more than one point. A deterministic tie-break converts that set-valued choice into one output. The infimum $\inf_{z \in D} E(z)$ is the greatest lower bound and need not be attained; the minimum is an attained infimum. Likewise, sup is the least upper bound.

A function E is *proper* in this monograph when it never takes $-\infty$ and is finite at at least one point. This is the standard extended-real optimization convention relevant to the existence theorems.

Probability used by Scout coverage

For a finite nonempty set D, write

$$\Delta(D) = \left\{ p : D \to [0,1] : \sum_{d \in D} p(d) = 1 \right\}.$$

Thus $\Delta(D)$ is the set of probability distributions on D. A stochastic policy maps its available information to an element of such a simplex. A deterministic policy is included as the special case that assigns probability one to a single action.

For events A and B with $\Pr(B) > 0$,

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}.$$

The union bound,

$$\Pr\left(\bigcup_{i=1}^{L} A_i \right) \le \sum_{i=1}^{L} \Pr(A_i),$$

does not require independence. The stochastic-completeness theorem also uses conditional expectation: if $\mathcal{H}$ records the current history and A is already determined by that history, then

$$\Pr(A \cap B) = \mathbb{E}[\mathbf{1}_A \Pr(B \mid \mathcal{H})].$$

This identity is the tower property applied to event indicators.

Optional topology for one existence theorem

A metric d measures distance and satisfies nonnegativity, identity of indiscernibles, symmetry, and the triangle inequality. A sequence z_n converges to z when $d(z_n, z) \to 0$. A metric space is compact when every sequence has a convergent subsequence whose limit remains in the space. A function E is lower semicontinuous when

$$z_n \to z \quad \Longrightarrow \quad E(z) \le \liminf_{n \to \infty} E(z_n).$$

These sequential formulations are sufficient for the compact metric theorem proved later.

State machines, search trees, and verifiers

A transition relation specifies which state-action-state triples are allowed. It need not be a deterministic function: the same state and action may have several possible successors. A search tree is the unrolling of such choices from initial states. A node is a partial trajectory; an edge is a legal extension.

A verifier is mathematically a total Boolean function on its declared dependent domain. An executable instantiation supplies a terminating procedure that computes that function for a claimed answer and certificate. "Independent" means that final acceptance is not defined merely as agreement with the learned score that proposed the answer. It does not mean that the verifier is infallible. Domain soundness of the verifier must be proved or inherited from a trusted exact checker.

Four properties must not be conflated:

Soundness Every emitted answer is correct.

Completeness Every target answer in the stated class is eventually found under the stated search assumptions.

Optimality The selected feasible answer minimizes a named objective.

Empirical accuracy A measured fraction of examples was answered correctly under a named evaluation.

None of these properties implies all the others.

A complete two-variable example

Let $x_1, x_2 \in \{0, 1\}$ and impose two constraints:

$$c_1 : \ x_1 \neq x_2, \qquad c_2 : \ x_1 = 1.$$

Define

$$E_{\text{hard}}(x_1, x_2) = \mathbf{1}\{x_1 = x_2\} + \mathbf{1}\{x_1 \neq 1\}.$$

Evaluating every assignment gives

(x_1, x_2)	$\mathbf{1}\{x_1 = x_2\}$	$\mathbf{1}\{x_1 \neq 1\}$	E_{hard}
$(0, 0)$	1	1	2
$(0, 1)$	0	1	1
$(1, 0)$	0	0	0
$(1, 1)$	1	0	1

so $(1, 0)$ is the unique zero-energy assignment.

An AHR realization can make the three roles explicit. The Queen authorizes the subgoal

"assign x_1 and x_2," sets a two-step budget, and requires verification. The Scout proposes variable-value branches. The Worker turns a retained proposal into a partial and then complete assignment. The exact verifier evaluates c_1 and c_2. A learned soft energy may choose which branch to try first, but only the zero-hard-energy assignment may be emitted. If the budget expires before that assignment is verified, the output is $\perp$, not the best unverified guess.

Part I

Foundations

Chapter 1

Origin and Design Commitments

1.1 The synthesis

AHR began as a synthesis of four ideas:

1. **Hierarchical recurrence.** Slow strategic state modulates faster local computation, extending the two-timescale HRM idea to Queen, Worker, and Scout recurrence [18, 2].
2. **Energy-based reasoning.** Correctness and preference are expressed through low energy, with hard constraints represented by extended-real indicator energies [19, 20].
3. **Stigmergic shared memory.** A bounded pheromone tensor records and redistributes evidence through evaporation and deposition [21, 24, 25].
4. **Independent verification.** AHR may propose, rank, repair, and search, but an external domain verifier controls final emission [3].

These ideas are complementary. Hierarchy allocates computation; energy ranks trajectories; stigmergy carries distributed evidence; verification separates plausible reasoning from certified correctness.

1.2 Non-negotiable architecture

Queen, Worker, and Scout are not labels attached after the fact. They are coordinates of the state and action spaces, coupled by an explicit compatibility relation. Removing any coordinate changes the formal system.

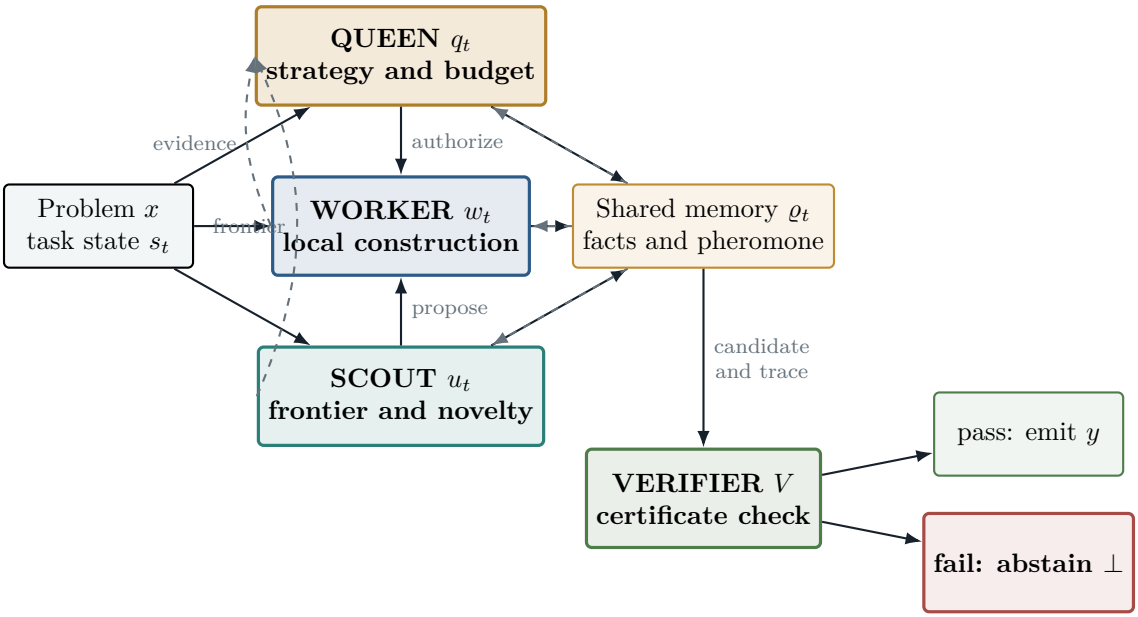


Figure 1.1: The AHR contract. Search is internal; final correctness is controlled by an external verifier.

1.3 Design requirements

The foundational theory must satisfy the following requirements.

- Correctness is defined externally by a semantic solution relation.
- The verifier is distinct from the semantic relation and from the learned energy.
- Hard constraints cannot be traded against soft preferences.
- The Queen may prune only under a stated safety condition if completeness is claimed.
- The Scout must receive an explicit coverage condition if eventual discovery is claimed.
- The Worker must receive a local completeness condition for the construction language in use.
- AHR returns $\perp$ when no verified bounded candidate is found.

Chapter 2

Semantic Contract

2.1 Problems, answers, and truth

Definition 2.1 (Semantic solution relation)**.** *A problem domain consists of a problem space X, an answer space Y, and a relation*

$$\mathrm{Sol} \subseteq X \times Y.$$

The statement $(x, y) \in \mathrm{Sol}$ means that y is correct for x under the intended external semantics of the domain.

The answer may be a value, assignment, proof object, plan, or certificate. The definition does not require uniqueness. For proof search it is often strongest to let Y contain proof objects rather than only theorem conclusions.

Definition 2.2 (Verifier)**.** *Let $\Omega(x)$ be the legal trajectory space for x. A verifier is a function on the dependent domain*

$$V : \{(x, \tau, y) : x \in X,\ \tau \in \Omega(x),\ y \in Y\} \longrightarrow \{0, 1\}.$$

$V(x, \tau, y) = 1$ means that the trajectory contains a certificate accepted for answer y.

Definition 2.3 (Verifier soundness)**.** *V is sound with respect to* Sol *when*

$$V(x, \tau, y) = 1 \quad \Longrightarrow \quad (x, y) \in \mathrm{Sol}$$

for every admissible triple (x, τ, y).

Verifier soundness is domain-specific. AHR cannot obtain it merely by architectural design. A proof assistant, exact CSP checker, or symbolic certificate checker must justify its own soundness.

2.2 Three correctness levels

Definition 2.4 (Verifier-relative soundness)**.** *A bounded solver* $\mathrm{AHR}_B : X \to Y \cup \{\perp\}$ *is verifier-sound when*

$$\mathrm{AHR}_B(x) = y \in Y \quad \Longrightarrow \quad \exists \tau \in \Omega(x) : V(x, \tau, y) = 1.$$

Definition 2.5 (Domain soundness)**.** AHR_B *is domain-sound when*

$$\mathrm{AHR}_B(x) = y \in Y \quad \Longrightarrow \quad (x, y) \in \mathrm{Sol}.$$

The logical chain is

$$\boxed{\text{AHR verifier-soundness} + \text{verifier domain-soundness} \Longrightarrow \text{AHR domain-soundness}.}$$

2.3 Abstention semantics

The symbol $\perp$ is a fresh symbol with $\perp \notin Y$ and means "no certified answer was emitted." It does not by itself distinguish impossibility, budget exhaustion, inadequate representation, incomplete verification, or search failure. An implementation may attach a reason code, but the mathematical output remains in $Y \cup \{\perp\}$.

Chapter 3

The AHR Mathematical Object

3.1 Primitive spaces

The finite base theory assumes all primitive sets are finite and nonempty unless stated otherwise. Let

$$X,\ Y,\ G,\ F,\ M,\ \mathcal{P},\ C$$

denote problems, answers, goals, facts, memory, partial constructions, and constraints. Define

$$S = G \times F \times M \times \mathcal{P} \times C, \qquad s = (g, f, m, p, c).$$

Let Q, W, U be the Queen, Worker, and Scout state spaces. The hierarchical state space is

$$H = Q \times W \times U \times S, \qquad h = (q, w, u, s).$$

Let A_Q, A_W, A_U be role action spaces and

$$A = A_Q \times A_W \times A_U, \qquad a = (a^Q, a^W, a^U).$$

3.2 Dynamics and trajectories

Definition 3.1 (Syntactic dynamics)**.** *The dynamic relation*

$$T_{\text{dyn}} \subseteq X \times H \times A \times H$$

contains syntactically possible state updates, including failed or incompatible attempts that may be useful for analysis.

Definition 3.2 (Initialization)**.** *An initialization correspondence $I : X \to 2^H$ assigns each problem a nonempty finite set $I(x)$ of allowed initial states.*

The strict AHR relation will be defined after compatibility:

$$T_{\text{AHR}} = \{(x, h, a, h') \in T_{\text{dyn}} : (q, w, u, s, a) \in \mathcal{A}_{\text{QWS}}\},$$

where $h = (q, w, u, s)$.

Definition 3.3 (Raw trajectory)**.** *For a problem x, let $\Omega_{\text{raw}}(x)$ be the set of all finite alternating strings*

$$\tau = (h_0, a_0, h_1, \ldots, a_{L-1}, h_L)$$

with $L \geq 0$, $h_0 \in I(x)$, $h_t \in H$, and $a_t \in A$, before any transition or compatibility test is imposed.

Definition 3.4 (Strict legal trajectory)**.** *A finite trajectory with transition length L is*

$$\tau = (h_0, a_0, h_1, \ldots, a_{L-1}, h_L).$$

It contains L actions and $L+1$ states. For $L = 0$ it consists only of an initial state h_0. It is strict-legal for x when $h_0 \in I(x)$ and

$$(x, h_t, a_t, h_{t+1}) \in T_{\text{AHR}} \quad \textit{for every } 0 \leq t < L.$$

The set of all such trajectories is $\Omega_{\text{AHR}}(x)$, abbreviated $\Omega(x)$.

The intermediate space $\Omega_{\text{dyn}}(x)$ consists of raw trajectories whose every transition belongs to T_{dyn}. It may contain QWS-incompatible attempts and is useful for training on negative examples. The three spaces satisfy

$$\Omega_{\text{AHR}}(x) \subseteq \Omega_{\text{dyn}}(x) \subseteq \Omega_{\text{raw}}(x).$$

This separation prevents an invalid attempt from being both "outside the declared domain" and "an in-domain example assigned infinite hard energy."

Definition 3.5 (Terminal extraction)**.** *The extraction map*

$$R : H \longrightarrow Y \cup \{\perp\}$$

returns the answer or certificate claimed by the final state.

3.3 Budget and policy

Let stepCost : $X \times H \times A \times H \to \mathbb{R}_{\geq 0}$. For a trajectory associated with problem x, define the additive, problem-indexed cost

$$\text{cost}_x(\tau) = \sum_{t=0}^{L-1} \text{stepCost}(x, h_t, a_t, h_{t+1}).$$

For budget $B \geq 0$,

$$\Omega_B(x) = \{\tau \in \Omega(x) : \text{cost}_x(\tau) \leq B\}.$$

The finite base theory additionally assumes that there is a constant $\varepsilon > 0$ such that

$$\text{stepCost}(x, h, a, h') \geq \varepsilon$$

for every strict-legal transition $(x, h, a, h') \in T_{\text{AHR}}$. Consequently a finite budget bounds transition length; the exact counting argument is proved in the existence chapter. A domain that needs zero-cost bookkeeping steps must either group them with a positive-cost transition

or add an explicit finite transition cap. Generalized infinite-space variants may replace this standing assumption by one of the later compactness or finite-sublevel hypotheses, but then minimizer existence must be checked separately.

The search policy is role-factored:

$$\Pi = (\Pi_Q, \Pi_W, \Pi_U).$$

One explicit Markov, autoregressive typing is

$$\begin{aligned}\Pi_Q &: X \times H \to \Delta(A_Q),\\ \Pi_U &: X \times H \times A_Q \to \Delta(A_U),\\ \Pi_W &: X \times H \times A_Q \times A_U \to \Delta(A_W).\end{aligned}$$

It induces the joint action probability

$$\Pr_{\Pi}(a^Q, a^W, a^U \mid x, h) = \Pi_Q(a^Q \mid x, h)\, \Pi_U(a^U \mid x, h, a^Q)\, \Pi_W(a^W \mid x, h, a^Q, a^U).$$

Successively summing this product over a^W, then a^U, then a^Q gives one because each conditional factor belongs to its displayed probability simplex. It is therefore a normalized joint distribution on A. The order of the three components in the tuple Π is only a naming convention; the displayed conditioning gives the common Queen–Scout–Worker sampling order. A history-dependent implementation either replaces H by the set of strict prefixes or augments H with all information needed to make the policy Markov. Policies may be masked to put zero mass on known-incompatible actions; regardless of masking, only actions that produce a strict transition belong to an AHR trajectory. The ideal AHR objective is independent of a particular policy. The policy determines how the ideal candidate set is enumerated or approximated.

3.4 Compact specification

Definition 3.6 (Finite strict AHR system)**.** *A finite strict AHR system is the tuple*

$$\begin{aligned}\mathfrak{A} = (&X, Y, \mathrm{Sol}, G, F, M, \mathcal{P}, C, S, Q, W, U, H,\\ &A_Q, A_W, A_U, A, \mathcal{K}, \mathcal{A}_{\mathrm{QWS}}, I, T_{\mathrm{dyn}}, T_{\mathrm{AHR}},\\ &\Omega_{\mathrm{raw}}, \Omega_{\mathrm{dyn}}, \Omega, R, V, E_{\mathrm{AHR}}, \mathrm{stepCost}, \{\mathrm{cost}_x\}_{x\in X}, B, \Pi).\end{aligned}$$

Every symbol is typed by the definitions above and below.

Chapter 4

Queen, Worker, and Scout as Primitive Structure

Consistently with the finite base theory, every component set introduced in this chapter is finite and nonempty unless explicitly stated otherwise. The products below are canonical uncompressed schemas. A domain instantiation may suppress coordinates whose sets are singletons, but it must state the resulting bijection; suppressing a singleton changes notation, not the mathematical state space. Non-singleton coordinates may be combined only when the required typed projections remain defined.

4.1 Queen state

Let Σ be a strategy set, $\mathcal{B}$ a budget-state set, $\mathcal{V}$ a verification-policy set, Θ an acceptance-threshold set, $\mathcal{Z}$ a stopping-rule set, and $\mathcal{A}_M$ an arbitration-memory set. Define

$$Q = \Sigma \times \mathcal{B} \times \mathcal{V} \times \Theta \times \mathcal{Z} \times \mathcal{A}_M.$$

A Queen state $q = (\sigma, b, \nu, \theta, z, \alpha)$ governs subgoals, resource allocation, pruning, verification, arbitration, acceptance, and stopping. The Queen does not need to construct domain proof steps.

4.2 Worker state

Let $\varnothing \neq G_W \subseteq G$ contain active subgoals, L local construction states, $\mathcal{R}$ domain rules, D derivation states, Λ local feedback, and Γ_P progress scores. Define

$$W = G_W \times L \times \mathcal{R} \times D \times \Lambda \times \Gamma_P.$$

A Worker state $w = (\widehat{g}, \ell, r, d, \lambda, \pi)$ records the authorized local obligation and its partial construction.

4.3 Scout state

Fix a finite nonempty branch universe $\mathcal{K}$. A branch may denote a hypothesis, rule, premise, variable-value choice, lemma, mutation, local plan, or search direction. Let $\mathcal{E}_K, \mathcal{N}_K, \mathcal{U}_K$ be finite nonempty grids of energy, novelty, and uncertainty values. Define

$$U = 2^{\mathcal{K}} \times 2^{\mathcal{K}} \times 2^{\mathcal{K}} \times \mathcal{E}_K^{\mathcal{K}} \times \mathcal{N}_K^{\mathcal{K}} \times \mathcal{U}_K^{\mathcal{K}}.$$

A Scout state is

$$u = (F_U, H_U, B_U, e_U, n_U, \xi_U),$$

where F_U is the active frontier, H_U stores hypotheses, and B_U contains unexplored branches. The maps e_U, n_U, and ξ_U store branch estimates. The Scout proposes; it does not certify correctness.

4.4 Nested time scales

For positive integers N, K, T, d, the original recurrent realization uses Queen updates $n = 0, \dots, N-1$, Worker updates $k = 0, \dots, K-1$ inside each Queen cycle, and Scout micro-steps $t = 0, \dots, T-1$ inside each Worker cycle. This is an AHR formalization of hierarchical temporal processing rather than a claim that biological and computational time scales are identical [18, 26]:

$$u_{n,k,t+1} = \mathrm{Step}_U(u_{n,k,t}; x, w_{n,k}, q_n, \varrho_{n,k,t}), \tag{4.1}$$

$$\varrho_{n,k,t+1} = \mathrm{Step}_\varrho(\varrho_{n,k,t}; x, q_n, w_{n,k}, u_{n,k,t}, s_{n,k,t}), \tag{4.2}$$

$$w_{n,k+1} = \mathrm{Step}_W(w_{n,k}; x, q_n, \mathcal{U}_{n,k}, \varrho_{n,k,T}), \tag{4.3}$$

$$q_{n+1} = \mathrm{Step}_Q(q_n; x, \mathcal{W}_n, \varrho_{n,K-1,T}). \tag{4.4}$$

The names $\mathrm{Step}_U, \mathrm{Step}_W, \mathrm{Step}_Q$ denote update functions and are deliberately distinct from the Scout frontier coordinate F_U. One fully typed choice is

$$\begin{aligned}
\mathrm{Step}_U &: U \times X \times W \times Q \times \mathbb{R}^d \to U, \\
\mathrm{Step}_\varrho &: \mathbb{R}^d \times X \times Q \times W \times U \times S \to \mathbb{R}^d, \\
\mathrm{Step}_W &: W \times X \times Q \times U^T \times \mathbb{R}^d \to W, \\
\mathrm{Step}_Q &: Q \times X \times W^K \times \mathbb{R}^d \to Q.
\end{aligned}$$

The task state $s_{n,k,t} \in S$ is supplied by the domain dynamics. Here take

$$\mathcal{U}_{n,k} = (u_{n,k,1}, \dots, u_{n,k,T}) \in U^T, \qquad \mathcal{W}_n = (w_{n,1}, \dots, w_{n,K}) \in W^K,$$

or replace either tuple by another explicitly typed summary of the same inner-loop states. An implementation also supplies

$$q_0 \in Q, \qquad \varrho_{0,0,0} \in \mathbb{R}^d, \qquad \mathrm{Reset}_W : Q \times X \to W, \qquad \mathrm{Reset}_U : Q \times W \times X \to U,$$

with

$$w_{n,0} = \mathrm{Reset}_W(q_n, x), \qquad u_{n,k,0} = \mathrm{Reset}_U(q_n, w_{n,k}, x).$$

Shared memory is carried across the nested-loop boundaries by

$$\varrho_{n,k+1,0} = \varrho_{n,k,T} \quad (0 \le k < K-1), \qquad \varrho_{n+1,0,0} = \varrho_{n,K-1,T} \quad (0 \le n < N-1).$$

The bounded evaporation–deposition rule in the next chapter is the canonical choice for Step_{ϱ}. These reset and carry equations define every displayed Queen, Worker, Scout, and memory index without an implicit off-by-one convention. The displayed $\mathbb{R}^d$ signatures describe the continuous recurrent realization. The finite base theory instead uses the finite encoded pheromone space and quantized update defined in the next chapter.

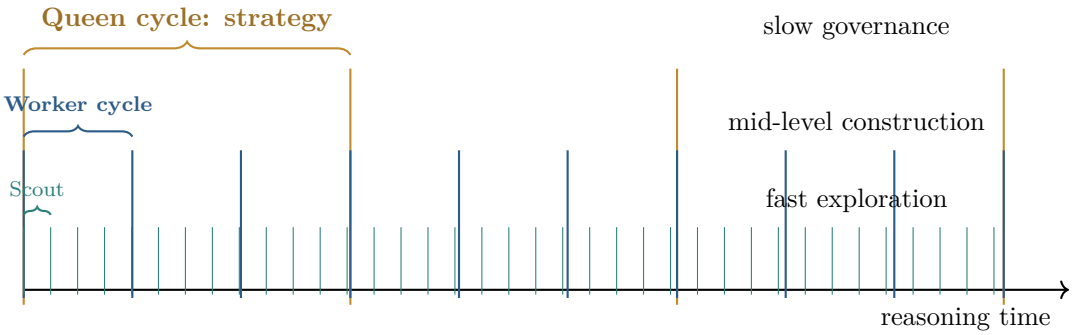


Figure 4.1: Three nested recurrence scales. The count NKT is computational depth, not parameter depth.

Chapter 5

The QWS Compatibility Relation

5.1 Auxiliary maps

The signature object of AHR is

$$\mathcal{A}_{\mathrm{QWS}} \subseteq Q \times W \times U \times S \times A.$$

To define it, fix the following maps and predicates:

$$\begin{aligned} &\mathrm{Auth} : Q \times S \to 2^{G}, && \mathrm{Rel} : Q \times S \to 2^{\mathcal{K}}, \\ &\mathrm{Adm} : Q \times S \times U \to 2^{\mathcal{K}}, && \mathrm{Rules} : Q \times S \times G \to 2^{\mathcal{R}}. \end{aligned}$$

Let

$$\mathcal{D}_{\mathrm{dec}} = \{\mathrm{undecided}, \mathrm{accept}, \mathrm{reject}, \mathrm{abstain}\}, \qquad \mathcal{V}_{\mathrm{stat}} = \{\mathrm{unchecked}, \mathrm{pass}, \mathrm{fail}\}.$$

The required action projections have types

$$\begin{aligned} \mathrm{Sel} &: A_Q \to 2^{G}, & \mathrm{Prune} &: A_Q \to 2^{\mathcal{K}}, & \mathrm{Verify} &: A_Q \to \{0,1\}, \\ \mathrm{Decision} &: A_Q \to \mathcal{D}_{\mathrm{dec}}, & \mathrm{Stop} &: A_Q \to \{0,1\}, & \mathrm{Sub} &: A_W \to 2^{G}, \\ \mathrm{Rule} &: A_W \to \mathcal{R} \cup \{\mathrm{none}\}, & \mathrm{Use} &: A_W \to 2^{\mathcal{K}}, & \mathrm{Prop} &: A_U \to 2^{\mathcal{K}}. \end{aligned}$$

The bit $\mathrm{Verify}(a^Q)$ requests an exact verifier call; it is not itself a claim that verification passed. A concrete T_{dyn} must give that bit operational meaning: a requested call records its returned status in the successor memory, while an unrequested call cannot manufacture a `pass` status. Moreover, $\mathrm{vstat}(m, w) = \mathrm{pass}$ may hold only when the recorded check applies to that exact Worker state or certificate; changing the checked construction invalidates or supersedes the pass. Clause QWS-5 below reads the status already present in the current state. Consequently the canonical transition model verifies first and accepts on a later transition. An atomic verify-and-accept variant must instead expose the prospective verifier result explicitly in its transition and compatibility semantics; the finite toy model later does this through its status v^+. Use the state projections

$$\widehat{g} : W \to G, \qquad \mathrm{Front} : U \to 2^{\mathcal{K}}, \qquad \mathrm{mem} : S \to M,$$

and a status map

$$\mathrm{vstat} : M \times W \to \mathcal{V}_{\mathrm{stat}}.$$

Finally, let $\mathrm{RequiresVerify} : Q \to \{0,1\}$ and type the evidence predicates as

$$\mathrm{PruneOK} : Q \times S \times U \times \mathcal{K} \to \{0,1\},$$
$$\mathrm{AcceptOK}, \mathrm{RejectOK}, \mathrm{FailureOK} : Q \times W \times U \times S \times A \to \{0,1\}.$$

Their domain-specific definitions make governance evidence-dependent. The relation $\mathcal{A}_{\mathrm{QWS}}$ checks whether the three roles make a coordinated request at the current state; it does not contain the successor h'. Accordingly, a faithful domain-specific T_{dyn} must also realize the declared action effects. For example, a branch named by $\mathrm{Prune}(a^Q)$ cannot remain in the retained successor frontier, selected subgoals must be reflected in the successor task/Worker state, and verifier status must obey the update contract above. A relation that ignores these effects is well-typed as a subset of $X \times H \times A \times H$ but is not a faithful AHR instantiation.

5.2 Exact compatibility clauses

For $q \in Q, w \in W, u \in U, s \in S, a \in A$, define

$$(q, w, u, s, a) \in \mathcal{A}_{\mathrm{QWS}}$$

if and only if all six clauses hold.

QWS-1: authorization

$$\widehat{g}(w) \in \mathrm{Auth}(q, s), \qquad \mathrm{Sel}(a^Q) \cup \mathrm{Sub}(a^W) \subseteq \mathrm{Auth}(q, s).$$

QWS-2: relevance

$\mathrm{Front}(u) \subseteq \mathrm{Rel}(q, s)$ and

$$\mathrm{Prop}(a^U) \setminus \mathrm{Prune}(a^Q) \subseteq \mathrm{Rel}(q, s).$$

The Scout may surface poor candidates only if they are prevented from entering the retained frontier.

QWS-3: admissible use

$$\mathrm{Use}(a^W) \subseteq (\mathrm{Front}(u) \cup \mathrm{Prop}(a^U)) \cap \mathrm{Adm}(q, s, u),$$

and

$$\mathrm{Rule}(a^W) = \mathrm{none} \quad \text{or} \quad \mathrm{Rule}(a^W) \in \mathrm{Rules}(q, s, \widehat{g}(w)).$$

QWS-4: evidence-bound governance

$$\mathrm{Prune}(a^Q) \subseteq \mathrm{Front}(u) \cup \mathrm{Prop}(a^U),$$

and for every $k \in \mathrm{Prune}(a^Q)$, $\mathrm{PruneOK}(q, s, u, k) = 1$. Moreover,

$$\mathrm{Decision}(a^Q) = \mathrm{accept} \Longrightarrow \mathrm{AcceptOK}(q, w, u, s, a) = 1,$$
$$\mathrm{Decision}(a^Q) = \mathrm{reject} \Longrightarrow \mathrm{RejectOK}(q, w, u, s, a) = 1.$$

QWS-5: verification policy

$$\left.\begin{aligned} \mathrm{RequiresVerify}(q) = 1, \\ \mathrm{Decision}(a^Q) = \mathrm{accept} \end{aligned}\right\} \Longrightarrow \mathrm{vstat}(\mathrm{mem}(s), w) = \mathrm{pass}.$$

QWS-6: justified stopping

$\mathrm{Stop}(a^Q) = 1$ implies either

$$\mathrm{Decision}(a^Q) = \mathrm{accept} \quad \text{and} \quad \mathrm{vstat}(\mathrm{mem}(s), w) = \mathrm{pass},$$

or

$$\mathrm{Decision}(a^Q) \in \{\mathrm{reject}, \mathrm{abstain}\} \quad \text{and} \quad \mathrm{FailureOK}(q, w, u, s, a) = 1.$$

These clauses coordinate roles but do not replace local rule checking or final verification. A role-compatible trajectory can still contain a domain-invalid proof attempt.

5.3 Hard and soft alignment

For $h = (q, w, u, s)$, define

$$\ell_{\mathrm{align}}(h, a) = \begin{cases} 0, & (q, w, u, s, a) \in \mathcal{A}_{\mathrm{QWS}}, \\ +\infty, & \text{otherwise.} \end{cases}$$

For $\tau = (h_0, a_0, \ldots, h_L)$,

$$E_{\mathrm{align}}(x, \tau) = \sum_{t=0}^{L-1} \ell_{\mathrm{align}}(h_t, a_t).$$

Let

$$\mathcal{Z}_{\mathrm{QWS}} = Q \times W \times U \times S \times A.$$

On finite encoded spaces, assume $\mathcal{A}_{\mathrm{QWS}} \neq \varnothing$, fix $\lambda_A \geq 0$, and equip $\mathcal{Z}_{\mathrm{QWS}}$ with a metric d. A relaxed version is

$$d_{\mathrm{align}}(z, \mathcal{A}_{\mathrm{QWS}}) = \min_{z' \in \mathcal{A}_{\mathrm{QWS}}} d(z, z'), \qquad E^{\mathrm{soft}}_{\mathrm{align}}(\tau) = \lambda_A \sum_t d_{\mathrm{align}}(z_t, \mathcal{A}_{\mathrm{QWS}}).$$

Here $z_t = (q_t, w_t, u_t, s_t, a_t)$ is the expanded state-action tuple, and the sum runs over $t = 0, \ldots, L-1$. The hard relation defines AHR. The soft distance is a learning or repair surrogate.

Chapter 6

Stigmergic Memory and Energy-Based Dynamics

6.1 Pheromone state

Pheromone memory is part of the AHR state, not an untracked external variable. An instantiation that uses it chooses a memory space M with a distinguished projection

$$\mathrm{pher} : M \to \mathcal{P}_\varrho, \qquad \varrho_t = \mathrm{pher}(\mathrm{mem}(s_t)).$$

The transition relation T_{dyn} must update that memory coordinate consistently with the recurrence below. Fix an integer $d \geq 1$ and scalars $\varrho_{\min}, \varrho_{\max} \in \mathbb{R}$ satisfying $\varrho_{\min} \leq \varrho_{\max}$. Fix $0 < \rho \leq 1$ and deposits $\Delta\varrho_t \in \mathbb{R}^d$.

There are two precise variants. In the continuous differentiable realization, take $\mathcal{P}_\varrho \subseteq \mathbb{R}^d$. In the finite base theory, choose a nonempty finite encoded grid

$$\mathcal{P}_\varrho^{\mathrm{fin}} \subseteq [\varrho_{\min}, \varrho_{\max}]^d$$

and a deterministic quantizer

$$\mathsf{Q}_\varrho : [\varrho_{\min}, \varrho_{\max}]^d \to \mathcal{P}_\varrho^{\mathrm{fin}}.$$

The finite update is

$$\varrho_{t+1} = \mathsf{Q}_\varrho\Big(\mathrm{Clamp}_{[\varrho_{\min},\varrho_{\max}]}((1-\rho)\varrho_t + \Delta\varrho_t)\Big).$$

Because its range is finite, placing this coordinate inside M preserves the finiteness of H. The continuous model below omits Q_ϱ; it belongs to the generalized, potentially infinite-state theory, so finite argmin existence cannot be invoked for it without a separate existence argument.

For the continuous analysis, let $\varrho_t \in \mathbb{R}^d$ be the task-dependent shared memory and apply the clamp coordinatewise. Coordinates may index graph edges, cell-value assignments, proof actions, facts, rules, or factor-graph messages. The following bounded evaporation-deposition form is adapted from ant-colony and MAX–MIN pheromone dynamics, while its differentiable

task-dependent deposit is specific to AHR [21, 22, 23, 24]:

$$\varrho_{t+1} = \mathrm{Clamp}_{[\varrho_{\min},\varrho_{\max}]}((1-\rho)\varrho_t + \Delta\varrho_t), \qquad 0 < \rho \le 1.$$

The deposit map has type

$$D_\theta : Q \times W \times U \times S \times X \to \mathbb{R}^d, \qquad \Delta\varrho_t = D_\theta(q_t, w_t, u_t, s_t, x),$$

and may be hand-defined or learned. A low-rank realization is available when $d = d_1 d_2$ and the vector is reshaped as $\mathrm{mat}(\varrho) \in \mathbb{R}^{d_1 \times d_2}$: one may parameterize the pre-clamp field or the deposit as $UV^\top$ with $U \in \mathbb{R}^{d_1 \times r}$ and $V \in \mathbb{R}^{d_2 \times r}$. This reduces storage when $r(d_1 + d_2) < d_1 d_2$. An arbitrary coordinatewise clamp of a rank-r matrix need not have rank at most r. Therefore an implementation that stores the *clamped state itself* in low-rank form must either recompress it (generally approximately) or prove that its chosen update preserves the rank constraint. The low-rank parameterization is not asserted to be exactly closed under the canonical clamp.

The exact linear expansion and variational formulas below concern this continuous analytical model. A quantized finite implementation remains bounded because its state lies in $\mathcal{P}_\varrho^{\mathrm{fin}}$, but it does not satisfy the unclamped linear expansion unless the quantizer acts as the identity along the trajectory.

Lemma 6.1 (Unclamped bound)**.** *Suppose $D \ge 0$, $\widetilde{\varrho}_{t+1} = (1-\rho)\widetilde{\varrho}_t + \Delta\varrho_t$, and $\|\Delta\varrho_t\|_\infty \le D$. Then*

$$\|\widetilde{\varrho}_t\|_\infty \le (1-\rho)^t \|\widetilde{\varrho}_0\|_\infty + \frac{D}{\rho}(1-(1-\rho)^t).$$

Proof. Set $a_t = \|\widetilde{\varrho}_t\|_\infty$. Since $0 < \rho \le 1$, the triangle inequality and absolute homogeneity give

$$\begin{aligned} a_{t+1} &= \|(1-\rho)\widetilde{\varrho}_t + \Delta\varrho_t\|_\infty \\ &\le (1-\rho)\|\widetilde{\varrho}_t\|_\infty + \|\Delta\varrho_t\|_\infty \\ &\le (1-\rho)a_t + D. \end{aligned}$$

We prove by induction that

$$a_t \le (1-\rho)^t a_0 + D\sum_{j=0}^{t-1}(1-\rho)^j.$$

The assertion is equality at $t = 0$, with the empty sum equal to zero. If it holds at t, then

$$\begin{aligned} a_{t+1} &\le (1-\rho)\left((1-\rho)^t a_0 + D\sum_{j=0}^{t-1}(1-\rho)^j\right) + D \\ &= (1-\rho)^{t+1} a_0 + D\sum_{j=1}^{t}(1-\rho)^j + D \\ &= (1-\rho)^{t+1} a_0 + D\sum_{j=0}^{t}(1-\rho)^j. \end{aligned}$$

Thus the induction is complete. The finite geometric sum is

$$\sum_{j=0}^{t-1}(1-\rho)^j = \frac{1-(1-\rho)^t}{1-(1-\rho)} = \frac{1-(1-\rho)^t}{\rho},$$

for $t \geq 1$, which yields the stated bound. The already-checked case $t = 0$ has an empty sum and does not require the geometric-sum identity. □

Corollary 6.2 (Steady-state and clamped bounds)**.** *Under the hypotheses of the preceding lemma, suppose the clamp endpoints satisfy*

$$\varrho_{\min} \leq \varrho_{\max}.$$

Then

$$\limsup_{t\to\infty} \|\widetilde{\varrho}_t\|_\infty \leq \frac{D}{\rho}.$$

Under the clamped update, every coordinate lies in $[\varrho_{\min}, \varrho_{\max}]$ *for every* $t \geq 1$*; if* ϱ_0 *already lies in this box, the statement holds for every* $t \geq 0$*.*

Proof. Because $0 \leq 1 - \rho < 1$, one has $(1-\rho)^t \to 0$. Taking the limit superior in the lemma's bound gives

$$\limsup_{t\to\infty} \|\widetilde{\varrho}_t\|_\infty \leq 0 + \frac{D}{\rho}(1-0) = \frac{D}{\rho}.$$

For the clamped recurrence, scalar clamping is defined by

$$\mathrm{Clamp}_{[\varrho_{\min},\varrho_{\max}]}(r) = \min\{\varrho_{\max}, \max\{\varrho_{\min}, r\}\}.$$

Its value is always in $[\varrho_{\min}, \varrho_{\max}]$. Applying it coordinatewise therefore puts ϱ_{t+1} in the box for every $t \geq 0$, proving the claim from time 1 onward. If ϱ_0 is also in the box, the claim includes $t = 0$. □

6.2 Exponential memory

Proposition 6.3 (Exact exponential-memory expansion)**.** *For the unclamped recurrence and every integer* $t \geq 0$*,*

$$\widetilde{\varrho}_t = (1-\rho)^t \widetilde{\varrho}_0 + \sum_{j=0}^{t-1} (1-\rho)^{t-1-j} \Delta\varrho_j.$$

Proof. At $t = 0$, the sum is empty and the right side equals $\widetilde{\varrho}_0$. Assume the formula holds at time t. Substitution into the recurrence gives

$$\begin{aligned}
\widetilde{\varrho}_{t+1} &= (1-\rho)\widetilde{\varrho}_t + \Delta\varrho_t \\
&= (1-\rho)\left((1-\rho)^t \widetilde{\varrho}_0 + \sum_{j=0}^{t-1} (1-\rho)^{t-1-j} \Delta\varrho_j\right) + \Delta\varrho_t \\
&= (1-\rho)^{t+1} \widetilde{\varrho}_0 + \sum_{j=0}^{t-1} (1-\rho)^{t-j} \Delta\varrho_j + (1-\rho)^0 \Delta\varrho_t \\
&= (1-\rho)^{t+1} \widetilde{\varrho}_0 + \sum_{j=0}^{t} (1-\rho)^{t-j} \Delta\varrho_j,
\end{aligned}$$

which is the required formula at $t + 1$. □

Thus pheromone is an exponentially weighted history of deposits. Evaporation sets the memory horizon, while clamping prevents unbounded accumulation.

6.3 Pheromone as a variational prior

Let $n \geq 1$, let the action index set be $\{1, \ldots, n\}$, let $\ell_j \in \mathbb{R}$, let $\kappa \in \mathbb{R}$ (normally $\kappa > 0$), and let

$$\pi_j = \frac{e^{\kappa \varrho_j}}{\sum_{i=1}^n e^{\kappa \varrho_i}} > 0.$$

On the probability simplex

$$\Delta_n = \left\{ p \in [0,1]^n : \sum_{j=1}^n p_j = 1 \right\},$$

use the convention $0 \log(0/\pi_j) = 0$ and consider

$$p^* \in \arg\min_{p \in \Delta_n} \left\{ -\sum_{j=1}^n p_j \ell_j + \mathrm{KL}(p \| \pi) \right\}, \qquad \mathrm{KL}(p\|\pi) = \sum_{j=1}^n p_j \log \frac{p_j}{\pi_j}.$$

Throughout this section, log is the natural logarithm. The score ℓ_j is a log-preference: increasing it increases the preferred probability of action j.

Proposition 6.4 (Log-prior equivalence)**.** *The unique minimizer is*

$$p_j^* = \frac{\pi_j e^{\ell_j}}{\sum_{i=1}^n \pi_i e^{\ell_i}} = \frac{\exp(\ell_j + \kappa \varrho_j)}{\sum_{i=1}^n \exp(\ell_i + \kappa \varrho_i)}.$$

Proof. To identify the only possible interior stationary point, introduce a multiplier λ for $\sum_j p_j = 1$. Differentiating the Lagrangian coordinatewise gives

$$0 = -\ell_j + \frac{\partial}{\partial p_j} \left(p_j \log \frac{p_j}{\pi_j} \right) + \lambda = -\ell_j + \log \frac{p_j}{\pi_j} + 1 + \lambda.$$

Hence

$$\log \frac{p_j}{\pi_j} = \ell_j - 1 - \lambda, \qquad p_j = \pi_j e^{\ell_j} e^{-1-\lambda}.$$

Summing over j and using $\sum_j p_j = 1$ yields

$$1 = e^{-1-\lambda} \sum_{i=1}^n \pi_i e^{\ell_i}, \qquad e^{-1-\lambda} = \left(\sum_{i=1}^n \pi_i e^{\ell_i} \right)^{-1}.$$

This gives the first displayed formula. Substituting the normalized pheromone prior cancels its common denominator and gives the second.

It remains to prove global optimality and uniqueness, including boundary points. Put

$$Z = \sum_{i=1}^n \pi_i e^{\ell_i}, \qquad r_j = \frac{\pi_j e^{\ell_j}}{Z}.$$

Every r_j is positive and $\sum_j r_j = 1$. For arbitrary $p \in \Delta_n$,

$$\begin{aligned} -\sum_j p_j \ell_j + \mathrm{KL}(p\|\pi) &= \sum_j p_j \left(\log \frac{p_j}{\pi_j} - \ell_j \right) \\ &= \sum_j p_j \log \frac{p_j}{\pi_j e^{\ell_j}} \\ &= \sum_j p_j \log \frac{p_j}{Z r_j} \\ &= \sum_j p_j \log \frac{p_j}{r_j} - \log Z \sum_j p_j \\ &= \mathrm{KL}(p\|r) - \log Z. \end{aligned}$$

For completeness, Gibbs' inequality follows here from $\log u \le u - 1$. Let $S = \{j : p_j > 0\}$. Then

$$\begin{aligned} \mathrm{KL}(p\|r) &= \sum_{j \in S} -p_j \log \frac{r_j}{p_j} \\ &\ge \sum_{j \in S} -p_j \left(\frac{r_j}{p_j} - 1 \right) \\ &= \sum_{j \in S} (p_j - r_j) = 1 - \sum_{j \in S} r_j \ge 0. \end{aligned}$$

Equality in the last inequality requires $S = \{1, \dots, n\}$ because every $r_j > 0$. Equality in $\log u \le u - 1$ then requires $r_j/p_j = 1$ for every j, so $p = r$. Conversely, $p = r$ plainly gives zero. Therefore $\mathrm{KL}(p\|r) \ge 0$ with equality if and only if $p = r$, and $r = p^*$ is the unique global minimizer. □

Example 6.5 (Two-action calculation). Take $\kappa = \log 2$, $\varrho_1 = 0$, and $\varrho_2 = 1$. Then

$$\pi_1 = \frac{1}{1+2} = \frac{1}{3}, \qquad \pi_2 = \frac{2}{1+2} = \frac{2}{3}.$$

The pheromone prior therefore favors action 2. Now take $\ell_1 = \log 3$ and $\ell_2 = 0$. The unnormalized posterior weights are

$$\pi_1 e^{\ell_1} = \frac{1}{3} \cdot 3 = 1, \qquad \pi_2 e^{\ell_2} = \frac{2}{3} \cdot 1 = \frac{2}{3}.$$

Their sum is $5/3$, so

$$p_1^* = \frac{1}{5/3} = \frac{3}{5}, \qquad p_2^* = \frac{2/3}{5/3} = \frac{2}{5}.$$

The current action score can therefore outweigh, but does not erase, the pheromone prior.

This gives stigmergy a precise EBM interpretation: pheromone is a structured prior, possibly learned, not merely a biological analogy.

Chapter 7

The AHR Energy Functional

7.1 Extended-real formulation

Let the common raw candidate domain be

$$\mathcal{D}_{\mathrm{raw}} = \{(x, \tau, y) : x \in X,\ \tau \in \Omega_{\mathrm{raw}}(x),\ y \in Y\}.$$

The energy is

$$E_{\mathrm{AHR}} : \mathcal{D}_{\mathrm{raw}} \to \mathbb{R} \cup \{+\infty\}.$$

Its canonical decomposition is

$$\boxed{E_{\mathrm{AHR}} = E_Q + E_W + E_U + E_{\mathrm{align}} + E_T + E_V + E_R + E_B.}$$

The role terms have the common type

$$E_Q, E_W, E_U : \mathcal{D}_{\mathrm{raw}} \to \mathbb{R}$$

and the following intended meanings:

$$\begin{aligned} E_Q &= \text{governance, verification timing, arbitration, and stopping cost,} \\ E_W &= \text{local construction error, malformedness, and lack of progress,} \\ E_U &= \text{irrelevance, duplication, unsupported exploration, and uncertainty cost.} \end{aligned}$$

The remaining terms enforce structure:

$$\begin{aligned} E_{\mathrm{align}} &= \text{QWS compatibility}, & E_T &= \text{transition legality}, \\ E_V &= \text{verification}, & E_R &= \text{terminal answer consistency}, \\ E_B &= \text{budget feasibility}. \end{aligned}$$

Whenever a term is naturally defined only on (x, τ), this monograph uses its canonical lift to $\mathcal{D}_{\mathrm{raw}}$,

$$\overline{F}(x, \tau, y) = F(x, \tau),$$

and then reuses the symbol F for the lift. In particular this convention applies to E_{align}, E_T, and E_B. Hence every summand in E_{AHR} has exactly the same domain.

7.2 Hard indicators

For $\tau = (h_0, a_0, \ldots, a_{L-1}, h_L) \in \Omega_{\text{raw}}(x)$, define the hard transition energy

$$E_T(x, \tau) = \sum_{t=0}^{L-1} \begin{cases} 0, & (x, h_t, a_t, h_{t+1}) \in T_{\text{dyn}}, \\ +\infty, & \text{otherwise.} \end{cases}$$

Extend the verifier to the raw domain by rejection:

$$V^{\text{ext}}(x, \tau, y) = \begin{cases} V(x, \tau, y), & \tau \in \Omega(x), \\ 0, & \tau \notin \Omega(x). \end{cases}$$

This piecewise definition calls V only in the branch on which its original dependent-domain requirement is satisfied. The hard verifier, extraction, and budget energies are

$$E_V(x, \tau, y) = \begin{cases} 0, & V^{\text{ext}}(x, \tau, y) = 1, \\ +\infty, & V^{\text{ext}}(x, \tau, y) = 0, \end{cases}$$

$$E_R(x, \tau, y) = \begin{cases} 0, & R(h_L) = y, \\ +\infty, & R(h_L) \neq y, \end{cases}$$

$$E_B(x, \tau) = \begin{cases} 0, & \text{cost}_x(\tau) \leq B, \\ +\infty, & \text{cost}_x(\tau) > B. \end{cases}$$

Likewise, $E_{\text{align}} = +\infty$ for a QWS-incompatible step. Therefore finite $E_T + E_{\text{align}}$ is equivalent to $\tau \in \Omega_{\text{AHR}}(x)$.

7.3 Hard constraints are not weights

A large finite penalty can be outweighed by enough soft reward. For example, if an invalid candidate has soft energy 0, a valid candidate has soft energy $M+1$, and invalidity incurs only the finite penalty M, then the invalid candidate has total energy $M < M+1$. Therefore foundational verification uses $+\infty$, or equivalently optimizes only over the feasible set. Learned confidence may rank candidates before verification but cannot override final rejection.

Proposition 7.1 (Equivalence of hard-energy and feasible-set inference)**.** *Let $\mathcal{D}$ be a candidate domain, let $E_{\text{soft}} : \mathcal{D} \to \mathbb{R}$ be finite-valued, and let $I_1, \ldots, I_m : \mathcal{D} \to \{0, +\infty\}$ be hard indicators. Define*

$$\mathcal{F} = \{z \in \mathcal{D} : I_j(z) = 0 \text{ for every } j\}, \qquad E_{\text{ext}}(z) = E_{\text{soft}}(z) + \sum_{j=1}^{m} I_j(z).$$

If $\mathcal{F} \neq \varnothing$, then

$$\arg\min_{z \in \mathcal{D}} E_{\text{ext}}(z) = \arg\min_{z \in \mathcal{F}} E_{\text{soft}}(z).$$

Proof. If $z \in \mathcal{F}$, every $I_j(z) = 0$, so $E_{\text{ext}}(z) = E_{\text{soft}}(z) \in \mathbb{R}$. If $z \notin \mathcal{F}$, at least one indicator equals $+\infty$. Since $E_{\text{soft}}(z)$ is finite,

$$E_{\text{ext}}(z) = E_{\text{soft}}(z) + (+\infty) = +\infty.$$

Because $\mathcal{F}$ is nonempty, some candidate has finite extended energy; therefore no point outside $\mathcal{F}$ can minimize E_{ext}. On $\mathcal{F}$, the two objectives agree pointwise. Thus a point minimizes E_{ext} over $\mathcal{D}$ if and only if it minimizes E_{soft} over $\mathcal{F}$, proving equality of the two argmin sets. □

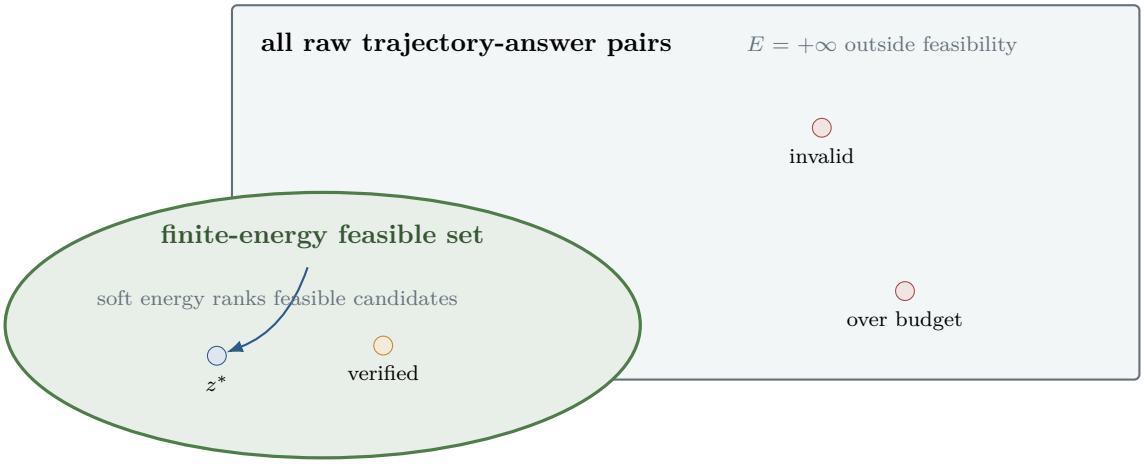


Figure 7.1: The extended-real EBM view. Hard constraints define feasibility; soft energy ranks only feasible candidates.

Chapter 8

Objective, Search, Repair, and Halting

8.1 Verified bounded candidates

For $x \in X$, define

$$\mathcal{C}_B(x) = \{(\tau, y) : \tau \in \Omega(x),\ R(h_L) = y,\ V(x, \tau, y) = 1,\ \mathrm{cost}_x(\tau) \le B\}.$$

Fix total orders on the finite primitive alphabets and use their induced shortlex order $\preceq$ on encoded trajectory–answer pairs: shorter encodings precede longer ones, and equal-length encodings are compared lexicographically.

Definition 8.1 (AHR output rule)**.** *If $\mathcal{C}_B(x) = \varnothing$, set $\mathrm{AHR}_B(x) = \perp$. Otherwise choose*

$$(\tau^*, y^*) \in \operatorname*{arg\,min}_{(\tau,y)\in\mathcal{C}_B(x)} E_{\mathrm{AHR}}(x, \tau, y)$$

as the $\preceq$-least minimizing pair, and return $\mathrm{AHR}_B(x) = y^*$.

The standing finite-base and positive-step-cost assumptions make $\mathcal{C}_B(x)$ finite. The existence chapter proves that the displayed argmin is therefore nonempty whenever $\mathcal{C}_B(x)$ is nonempty. In a generalized infinite-space model, this definition is used only after one of the stated existence conditions has been verified.

This is constrained EBM inference over trajectories. The preceding proposition makes precise the equivalence of the hard-energy and feasible-set formulations.

8.2 Role-factored search

At a partial trajectory $\bar{\tau}$, the Scout proposes branches, the Queen filters and allocates, and the Worker realizes local successors. For a nonempty finite available-action set $\mathcal{A}(h)$ and a finite-valued learned score $E_\theta : X \times H \times A \to \mathbb{R}$, a common induced distribution, for $a \in \mathcal{A}(h)$, is

$$p_\theta(a \mid h, x) = \frac{\exp[-E_\theta(x, h, a)]}{\sum_{a' \in \mathcal{A}(h)} \exp[-E_\theta(x, h, a')]},$$

with role factorization or autoregressive conditioning such as

$$p(a^Q, a^W, a^U \mid h, x) = p_Q(a^Q \mid h, x)p_U(a^U \mid a^Q, h, x)p_W(a^W \mid a^Q, a^U, h, x).$$

This factorization is a modeling choice; compatibility remains a hard relation.

8.3 Verifier-guided repair

Given a decoded candidate $y_0 \in Y$, a metric $d : Y \times Y \to \mathbb{R}_{\geq 0}$, and a hard candidate energy $E_{\text{hard}} : X \times Y \to \mathbb{R} \cup \{+\infty\}$, bounded repair searches

$$\mathcal{B}_r(y_0) = \{y \in Y : d(y, y_0) \leq r\}, \qquad y^* \in \operatorname*{arg\,min}_{y \in \mathcal{B}_r(y_0)} E_{\text{hard}}(x, y).$$

Here $r \geq 0$; in the finite base theory the ball is nonempty because it contains y_0, so the displayed argmin exists. The candidate is emitted only when a corresponding trajectory and certificate pass V. Repair is therefore a bounded extension of search, not a relaxation of correctness.

8.4 Adaptive computation

Fix a finite Queen-cycle horizon N. At decision stage $n \in \{0, \ldots, N-1\}$, let the Markov state be $z_n = (q_n, \varrho_n, s_n)$ and let $c_n \in \{\text{halt}, \text{continue}\}$. Let $\ell(z)$ be a named task loss and let $\text{Cons}(\varrho)$ be a named constraint-consistency score. A common reward for a continuing transition $z_n \to z_{n+1}$ is

$$r_n^{\text{cont}} = \ell(z_n) - \ell(z_{n+1}) + \mu\big(\text{Cons}(\varrho_{n+1}) - \text{Cons}(\varrho_n)\big) - \lambda,$$

where $\mu, \lambda \geq 0$. The task must separately specify the terminal reward for halting and the forced terminal condition at stage N. This makes halting an optimal-stopping problem related to adaptive recurrent compute and temporal abstraction [27, 28].

For a specified finite-horizon Markov transition law with integrable rewards, define $Q_n^*(z, c)$ as the optimal expected cumulative reward from stage n onward after taking action c in state z at that stage. Equivalently, one may augment z with the remaining horizon and use an unindexed Q^*. Because the action set has two elements, finite-horizon Bellman optimality gives

$$c_n^* \in \operatorname*{arg\,max}_{c \in \{\text{halt}, \text{continue}\}} Q_n^*(z_n, c).$$

This statement is exact but tautological; a useful theory must bound approximation error and relate the compute penalty λ to expected budget.

Part II

Guarantees

Guarantee Roadmap

The guarantees are conditional and modular. The following table names the extra premise that does the essential work in each result.

Table 8.1: Logical scope of the principal guarantees.

Result	Essential additional premises	What follows
Finite existence	Finite state/action sets, positive minimum step cost, finite budget	Finitely many bounded candidates and a selectable minimizer
Topological existence	Nonempty compact candidate space, proper lower-semicontinuous energy	At least one minimizer exists
Finite-sublevel existence	Proper energy and finite sublevel sets	At least one global minimizer exists
Domain soundness	AHR emits only verifier passes and the verifier is domain-sound	Every emitted answer is semantically correct
Margin recovery	A verified candidate exists and a positive score gap separates it from failures	Learned-score minimizers are verified on the visited set
Safe pruning	A true lower bound exceeds a finite incumbent energy	No tying or improving completion is removed
Deterministic completeness	Target representation, Scout coverage, safe retention, Worker realizability, fair scheduling, terminal verifier evaluation	The named finite target is reached and its verifier pass is observed
Probabilistic completeness	The same structure plus conditional per-revisit success lower bounds	Target-failure probability tends to zero under staged revisits
Role-work advantage	Fixed branching and cost model with $1 < rm < b$	A conditional asymptotic work ratio tends to zero

Chapter 9

Existence and Well-Posedness

Proposition 9.1 (Finite bounded candidate set)**.** *Assume H and A are finite, every legal transition costs at least $\varepsilon > 0$, trajectory cost is additive, and $B < \infty$. Then $\Omega_B(x)$ and $\mathcal{C}_B(x)$ are finite.*

Proof. For a trajectory with L transitions, put

$$c_t = \text{stepCost}(x, h_t, a_t, h_{t+1}) \qquad (0 \le t < L).$$

Its additive cost is $\sum_{t=0}^{L-1} c_t$. Since each $c_t \ge \varepsilon$,

$$B \ge \text{cost}_x(\tau) = \sum_{t=0}^{L-1} c_t \ge \sum_{t=0}^{L-1} \varepsilon = L\varepsilon.$$

Hence $L \le B/\varepsilon$, and because L is an integer, $L \le L_{\max} := \lfloor B/\varepsilon \rfloor$. For fixed L, there are at most

$$|H|^{L+1}|A|^L$$

alternating state-action strings $(h_0, a_0, \ldots, a_{L-1}, h_L)$. Consequently

$$|\Omega_B(x)| \le \sum_{L=0}^{L_{\max}} |H|^{L+1}|A|^L < \infty.$$

Legal-initial-state and transition restrictions can only reduce this count. Finally, $\mathcal{C}_B(x)$ is a subset of $\Omega_B(x) \times Y$ whose answer is fixed by $y = R(h_L)$. Thus each trajectory contributes at most one answer and $|\mathcal{C}_B(x)| \le |\Omega_B(x)| < \infty$. □

Corollary 9.2 (Finite argmin existence)**.** *If $\mathcal{C}_B(x)$ is nonempty under the preceding assumptions, then the AHR argmin is nonempty and deterministic tie-breaking makes* $\text{AHR}_B(x)$ *single-valued.*

Proof. Write $\mathcal{C}_B(x) = \{z_1, \ldots, z_N\}$ with $N \ge 1$. The image

$$\{E_{\text{AHR}}(x, z_i) : 1 \le i \le N\} \subseteq \mathbb{R}$$

is finite and nonempty: membership in $\mathcal{C}_B(x)$ makes every hard term zero, while E_Q, E_W, E_U

are finite-valued by definition. A finite nonempty set of real numbers has a least value. Hence

$$\arg\min_{z\in\mathcal{C}_B(x)} E_{\mathrm{AHR}}(x,z) \neq \varnothing.$$

A fixed total order on the encoded finite set has a unique least element in this nonempty argmin set. The tie-breaking operator selects that element, so the returned answer is single-valued. □

Theorem 9.3 (Topological existence)**.** *Let $\mathcal{C}_B(x)$ be a nonempty compact metric space. If $E_{\mathrm{AHR}}(x,\cdot) : \mathcal{C}_B(x) \to \mathbb{R}\cup\{+\infty\}$ is proper (finite at one or more points) and lower semicontinuous, then it attains its minimum.*

Proof. Abbreviate $\mathcal{C}_B(x)$ by K and the energy by E. Properness supplies $z_0 \in K$ with $E(z_0) < +\infty$, and E never takes the value $-\infty$. Let

$$m = \inf_{z\in K} E(z) \leq E(z_0) < +\infty.$$

First, $m > -\infty$. Otherwise, for every integer $n \geq 1$ there would be $z_n \in K$ with $E(z_n) \leq -n$. Compactness of the metric space gives a convergent subsequence $z_{n_k} \to z^* \in K$. Lower semicontinuity would imply

$$E(z^*) \leq \liminf_{k\to\infty} E(z_{n_k}) = -\infty,$$

because the strictly increasing subsequence indices satisfy $n_k \to \infty$ and hence $E(z_{n_k}) \leq -n_k \to -\infty$. The displayed inequality then forces $E(z^*) = -\infty$ in the extended order, but $-\infty$ is not in the codomain $\mathbb{R}\cup\{+\infty\}$. Equivalently, no permitted value of $E(z^*)$ can be less than or equal to $-\infty$. This is a contradiction.

Thus $m \in \mathbb{R}$. By the definition of infimum, for every $n \geq 1$ choose $z_n \in K$ satisfying

$$m \leq E(z_n) < m + \frac{1}{n}.$$

Compactness again yields a subsequence $z_{n_k} \to \widehat{z} \in K$. The bounds on the minimizing sequence imply

$$m \leq \liminf_{k\to\infty} E(z_{n_k}) \leq \limsup_{k\to\infty} E(z_{n_k}) \leq \lim_{k\to\infty} \left(m + \frac{1}{n_k}\right) = m.$$

Thus both the limit inferior and limit superior equal m. Lower semicontinuity now gives

$$E(\widehat{z}) \leq \liminf_{k\to\infty} E(z_{n_k}) = m.$$

Since m is a lower bound, also $m \leq E(\widehat{z})$. Therefore $E(\widehat{z}) = m$, so $\widehat{z}$ is a minimizer. □

Proposition 9.4 (Finite-sublevel existence)**.** *Let $\mathcal{D}$ be any set and let $E : \mathcal{D} \to \mathbb{R}\cup\{+\infty\}$ be proper. If every finite sublevel set*

$$\{z \in \mathcal{D} : E(z) \leq c\}$$

is finite for every $c \in \mathbb{R}$, then E attains its minimum on $\mathcal{D}$.

Proof. Properness gives $z_0 \in \mathcal{D}$ with $c_0 := E(z_0) \in \mathbb{R}$. The sublevel set

$$S_0 = \{z \in \mathcal{D} : E(z) \leq c_0\}$$

is nonempty because it contains z_0, and it is finite by assumption. Hence there is $z^* \in S_0$ whose energy is least among the finitely many energies on S_0. If $z \in S_0$, then $E(z^*) \leq E(z)$. If $z \notin S_0$, then $E(z) > c_0 \geq E(z^*)$. Thus $E(z^*) \leq E(z)$ for every $z \in \mathcal{D}$, so z^* is a global minimizer. □

Chapter 10

Soundness and Abstention

Proposition 10.1 (Verifier-relative soundness)**.** *The AHR output rule is sound relative to V.*

Proof. Fix $x \in X$ and suppose $\mathrm{AHR}_B(x) = y \in Y$. The output rule returns an element of Y only in the nonempty case, so $\mathcal{C}_B(x) \neq \varnothing$ and there is a selected pair $(\tau^*, y^*) \in \mathcal{C}_B(x)$ with $y = y^*$. By the defining membership conditions of $\mathcal{C}_B(x)$,

$$V(x, \tau^*, y^*) = 1.$$

Substituting $y = y^*$ gives $V(x, \tau^*, y) = 1$, which is verifier-relative soundness. □

Theorem 10.2 (Domain soundness transfer)**.** *If V is sound with respect to* Sol*, then* AHR_B *is domain-sound.*

Proof. Fix $x \in X$ and $y \in Y$ and assume $\mathrm{AHR}_B(x) = y$. Verifier-relative soundness supplies a trajectory τ such that $V(x, \tau, y) = 1$. The assumed domain soundness of V is the implication

$$\forall x' \in X,\ \forall \tau' \in \Omega(x'),\ \forall y' \in Y, \qquad V(x', \tau', y') = 1 \Longrightarrow (x', y') \in \mathrm{Sol}.$$

Instantiate it with $(x', \tau', y') = (x, \tau, y)$ to obtain $(x, y) \in \mathrm{Sol}$. □

Proposition 10.3 (Abstention safety)**.** *If $\mathcal{C}_B(x) = \varnothing$, AHR emits no element of Y. Therefore lack of a verified bounded candidate cannot produce an unverified answer.*

Proof. If $\mathcal{C}_B(x) = \varnothing$, the second branch of the output definition applies and gives $\mathrm{AHR}_B(x) = \perp$. By construction $\perp \notin Y$. Hence there is no $y \in Y$ satisfying $\mathrm{AHR}_B(x) = y$, so in particular no unverified element of Y is emitted. □

Abstention safety does not imply calibrated abstention. A system may abstain too often because the verifier is incomplete, the search policy is weak, or the representation omits valid solutions.

Chapter 11

Alignment, Margins, and Robustness

Theorem 11.1 (Alignment coherence)**.** *Let $\tau = (h_0, a_0, \dots, a_{L-1}, h_L)$ with $h_t = (q_t, w_t, u_t, s_t)$. If $E_{\text{align}}(x, \tau) = 0$, then*

$$(q_t, w_t, u_t, s_t, a_t) \in \mathcal{A}_{\text{QWS}} \qquad \textit{for every } t \in \{0, \dots, L-1\}.$$

Proof. Write $\tau = (h_0, a_0, \dots, h_L)$. Every summand $\ell_{\text{align}}(h_t, a_t)$ belongs to $\{0, +\infty\}$. If some step were outside $\mathcal{A}_{\text{QWS}}$, its summand would be $+\infty$, and the finite sum would equal $+\infty$, not 0. Therefore every summand is 0. By the definition of the local loss, $(q_t, w_t, u_t, s_t, a_t) \in \mathcal{A}_{\text{QWS}}$ for every $t \in \{0, \dots, L-1\}$. □

11.1 Two margin statements

Fix a problem instance x. Let $\mathcal{D}$ be a nonempty finite set of well-typed pairs $z = (\tau, y)$ with $\tau \in \Omega(x)$ and $y \in Y$. Abbreviate $V_x(z) = V(x, \tau, y)$, and let $E_x : \mathcal{D} \to \mathbb{R}$ be the finite-valued score used in learned ranking. Define

$$\mathcal{D}^+ = \{z \in \mathcal{D} : V_x(z) = 1\}, \qquad \mathcal{D}^- = \{z \in \mathcal{D} : V_x(z) = 0\}.$$

Here and below $\inf \varnothing := +\infty$.

Theorem 11.2 (Strong energy-margin recovery)**.** *If $\mathcal{D}^+ \neq \varnothing$ and*

$$\sup_{z^+ \in \mathcal{D}^+} E_x(z^+) + \gamma \leq \inf_{z^- \in \mathcal{D}^-} E_x(z^-)$$

for some $\gamma > 0$, then every verified candidate has lower energy than every invalid candidate; hence every minimizer is verified.

Proof. If $\mathcal{D}^- = \varnothing$, every candidate is verified and both conclusions are immediate. Assume henceforth that $\mathcal{D}^- \neq \varnothing$. Choose arbitrary $z^+ \in \mathcal{D}^+$ and $z^- \in \mathcal{D}^-$. By the definitions of supremum and infimum,

$$E_x(z^+) \leq \sup_{\widetilde{z}^+ \in \mathcal{D}^+} E_x(\widetilde{z}^+)$$

and

$$\inf_{\widetilde{z}^- \in \mathcal{D}^-} E_x(\widetilde{z}^-) \le E_x(z^-).$$

Combining these inequalities with the assumed margin gives

$$E_x(z^+) + \gamma \le \sup_{\widetilde{z}^+ \in \mathcal{D}^+} E_x(\widetilde{z}^+) + \gamma \le \inf_{\widetilde{z}^- \in \mathcal{D}^-} E_x(\widetilde{z}^-) \le E_x(z^-).$$

Since $\gamma > 0$, $E_x(z^+) < E_x(z^-)$. Thus no invalid candidate can be a global minimizer because $\mathcal{D}^+$ is nonempty. A global minimizer exists because $\mathcal{D}$ is finite and nonempty and E_x is real-valued. Therefore every global minimizer is verified. □

Theorem 11.3 (Weak energy-margin recovery)**.** *If $\mathcal{D}^+ \neq \varnothing$ and, for some $\gamma > 0$,*

$$\min_{z^+ \in \mathcal{D}^+} E_x(z^+) + \gamma \le \inf_{z^- \in \mathcal{D}^-} E_x(z^-),$$

then every global minimizer of E_x on $\mathcal{D}$ is verified.

Proof. If $\mathcal{D}^- = \varnothing$, every candidate is verified and the conclusion is immediate. Assume henceforth that $\mathcal{D}^- \neq \varnothing$. Because $\mathcal{D}^+$ is finite and nonempty, choose $z^+_{\min} \in \mathcal{D}^+$ satisfying

$$E_x(z^+_{\min}) = \min_{z^+ \in \mathcal{D}^+} E_x(z^+).$$

Let z^* be any global minimizer on the nonempty finite set $\mathcal{D}$. Suppose for contradiction that $z^* \in \mathcal{D}^-$. Then

$$\begin{aligned} E_x(z^*) &\ge \inf_{z^- \in \mathcal{D}^-} E_x(z^-) \\ &\ge E_x(z^+_{\min}) + \gamma \\ &> E_x(z^+_{\min}), \end{aligned}$$

where the final inequality uses $\gamma > 0$. This contradicts global minimality because $z^+_{\min} \in \mathcal{D}$. Hence $z^* \in \mathcal{D}^+$. Since z^* was arbitrary, every global minimizer is verified. □

The strong condition separates *every* verified candidate from every invalid one. The weak condition separates only the best verified candidate, which is already sufficient to exclude invalid global minimizers.

11.2 Perturbation stability

Theorem 11.4 (Stable minimizer under energy perturbation)**.** *Let $\mathcal{F}$ be a nonempty finite feasible set, let $E, \widehat{E} : \mathcal{F} \to \mathbb{R}$, and let z^* be the unique minimizer of E on $\mathcal{F}$. Suppose, for some $\gamma > 0$,*

$$E(z) \ge E(z^*) + \gamma \qquad \text{for every } z \in \mathcal{F} \setminus \{z^*\}.$$

If

$$\sup_{z \in \mathcal{F}} |\widehat{E}(z) - E(z)| < \frac{\gamma}{2},$$

then z^* is also the unique minimizer of $\widehat{E}$ on $\mathcal{F}$.

Proof. For $z \in \mathcal{F} \setminus \{z^*\}$,

$$\begin{aligned}\widehat{E}(z) &> E(z) - \frac{\gamma}{2} \\ &\geq E(z^*) + \gamma - \frac{\gamma}{2} \\ &= E(z^*) + \frac{\gamma}{2} \\ &> \widehat{E}(z^*).\end{aligned}$$

The first and last strict inequalities use $|\widehat{E}(\cdot) - E(\cdot)| < \gamma/2$, and the middle inequality uses the assumed gap. Thus every $z \neq z^*$ has strictly larger perturbed energy, so z^* remains the unique minimizer. □

This theorem gives a precise target for learned energies: approximation error must be smaller than half the decision gap on the candidate set of interest.

Chapter 12

Safe Pruning and Completeness

12.1 Prefix notation

For trajectories $\bar{\tau}$ and τ, write $\bar{\tau} \preceq \tau$ when $\bar{\tau}$ is an exact prefix of τ. Define

$$\mathrm{Comp}_x(\bar{\tau}) = \{(\tau, y) : \bar{\tau} \preceq \tau,\ \tau \in \Omega(x),\ y \in Y\}.$$

Adopt the extended-real convention $\inf \varnothing = +\infty$. A lower bound $\underline{E}(x, \bar{\tau})$ is admissible when

$$\underline{E}(x, \bar{\tau}) \leq \inf_{(\tau,y)\in \mathrm{Comp}_x(\bar{\tau})} E_{\mathrm{AHR}}(x, \tau, y).$$

Theorem 12.1 (Safe Queen pruning)**.** *Let $E_{\mathrm{best}} \in \mathbb{R}$ be the energy of a verified feasible incumbent. If*

$$\underline{E}(x, \bar{\tau}) > E_{\mathrm{best}},$$

then pruning $\bar{\tau}$ cannot remove a completion that improves or ties the incumbent.

Proof. If $\mathrm{Comp}_x(\bar{\tau}) = \varnothing$, pruning removes no completion and is trivially safe. Otherwise, choose arbitrary $(\tau, y) \in \mathrm{Comp}_x(\bar{\tau})$. Admissibility gives

$$\underline{E}(x, \bar{\tau}) \leq \inf_{(\tau',y')\in \mathrm{Comp}_x(\bar{\tau})} E_{\mathrm{AHR}}(x, \tau', y') \leq E_{\mathrm{AHR}}(x, \tau, y).$$

Together with the pruning test,

$$E_{\mathrm{best}} < \underline{E}(x, \bar{\tau}) \leq E_{\mathrm{AHR}}(x, \tau, y).$$

Thus every completion is strictly worse than the incumbent. Since the completion was arbitrary, no improving or tying completion is removed. □

12.2 Deterministic coverage

For a target $\tau^* = (h_0, a_0, \ldots, h_L)$, let

$$\mathrm{Pref}(\tau^*) = \{(h_0), (h_0, a_0, h_1), \ldots, \tau^*\}.$$

Theorem 12.2 (Coverage completeness)**.** *Suppose* $(\tau^*, y^*) \in \mathcal{C}_B(x)$ *and:*

1. *at every proper prefix of* τ^**, the Scout eventually proposes every branch needed for the next target action;*
2. *the Queen never prunes a prefix of* τ^**;*
3. *the Worker can realize the required next local construction whenever it is authorized and supplied;*
4. *the scheduler eventually selects and authorizes every retained target extension;*
5. *the verifier is evaluated when the reached terminal target is considered for emission;*
6. $|\tau^*| < \infty$.

Then exhaustive AHR search eventually reaches τ^**, evaluates* $V(x, \tau^*, y^*) = 1$*, and finds a verified answer.*

Proof. Write

$$\tau^* = (h_0, a_0, h_1, \ldots, a_{L-1}, h_L)$$

with finite L. For $i \in \{0, \ldots, L\}$, let $P_i = (h_0, a_0, \ldots, h_i)$ be the prefix ending at h_i. We prove by induction on i that the search eventually reaches P_i.

The base case $P_0 = (h_0)$ is reached at initialization because $\tau^* \in \Omega(x)$ implies $h_0 \in I(x)$. Assume P_i is reached for some $i < L$. By assumption 1, a Scout eventually proposes the branch data needed for target action a_i. By assumption 2, the Queen retains P_i and its target extension. By assumption 4, the scheduler eventually selects and authorizes that retained extension. Assumption 3 then lets the Worker realize the required successor h_{i+1}, so the search reaches P_{i+1}.

Finite induction yields $P_L = \tau^*$. Since $(\tau^*, y^*) \in \mathcal{C}_B(x)$, one has $V(x, \tau^*, y^*) = 1$. Assumption 5 invokes that verifier on the reached target, so the search finds a verified answer. □

12.3 Stochastic coverage

Theorem 12.3 (Probabilistic completeness)**.** *Fix* $(\tau^*, y^*) \in \mathcal{C}_B(x)$*, where* τ^* *has* $L \geq 0$ *required extensions, and suppose a staged fair scheduler works through them in order. Conditional on extensions* $1, \ldots, i-1$ *having been realized, the scheduler gives the active target prefix up to* $n \geq 1$ *revisits, stopping that stage early if extension* i *is proposed. On its* j*th revisit, conditional on the complete preceding history and on failures at its preceding revisits, the Scout proposes extension* i *with probability at least* $p_i \in (0, 1]$*. Assume safe Queen retention, that the Worker realizes every proposed target extension, and that the verifier is evaluated on a reached terminal target. Then*

$$\Pr[\textit{target not reached}] \leq \sum_{i=1}^{L} (1 - p_i)^n,$$

which tends to zero as $n \to \infty$*. On the complementary event, evaluation returns* $V(x, \tau^*, y^*) = 1$.

Proof. If $L = 0$, the target is its initial state, so it is reached at initialization; the displayed sum is empty and equals zero. Verification then returns one because $(\tau^*, y^*) \in \mathcal{C}_B(x)$. Assume henceforth that $L \geq 1$.

Fix extension i. Let G_i be the event that extensions $1, \ldots, i-1$ have been realized, so stage i becomes active. Let $F_{i,j}$ denote failure to propose extension i on revisit j. Off histories on which that revisit is not reached, set $F_{i,j}$ to false; it is used below only together with the event that all earlier revisits failed. Define

$$A_{i,0} = G_i, \qquad A_{i,j} = G_i \cap \bigcap_{r=1}^{j} F_{i,r} \quad (1 \leq j \leq n).$$

The event $A_{i,j-1}$ is determined by the complete history $\mathcal{H}_{i,j-1}$ before revisit j. On that event the conditional failure bound is

$$\Pr(F_{i,j} \mid \mathcal{H}_{i,j-1}) \leq 1 - p_i \quad \text{almost surely.}$$

Using the tower property for event indicators gives

$$\begin{aligned}
\Pr(A_{i,j}) &= \mathbb{E}\Big[\mathbf{1}_{A_{i,j-1}} \mathbf{1}_{F_{i,j}}\Big] \\
&= \mathbb{E}\Big[\mathbf{1}_{A_{i,j-1}} \mathbb{E}[\mathbf{1}_{F_{i,j}} \mid \mathcal{H}_{i,j-1}]\Big] \\
&= \mathbb{E}\Big[\mathbf{1}_{A_{i,j-1}} \Pr(F_{i,j} \mid \mathcal{H}_{i,j-1})\Big] \\
&\leq (1 - p_i) \Pr(A_{i,j-1}).
\end{aligned}$$

Iterating this inequality from $j = 1$ to $j = n$ yields

$$\Pr(A_{i,n}) \leq (1 - p_i)^n \Pr(G_i) \leq (1 - p_i)^n.$$

No independence assumption has been made.

Under safe Queen retention, fair staging, and Worker realizability, failure to reach the full target means that some stage became active and then failed on all n allocated revisits. Thus

$$\{\text{target not reached}\} \subseteq \bigcup_{i=1}^{L} A_{i,n}.$$

The union bound now gives

$$\Pr[\text{target not reached}] \leq \sum_{i=1}^{L} \Pr(A_{i,n}) \leq \sum_{i=1}^{L} (1 - p_i)^n.$$

For each i, $0 \leq 1 - p_i < 1$, so $(1 - p_i)^n \to 0$. A finite sum of these terms also tends to zero. □

This theorem makes explicit what "the Scout explores enough" must mean. Completeness is conditional on representation, coverage, safe pruning, local realizability, and fair scheduling.

Chapter 13

Complexity, Training Depth, and Compute

13.1 Conditional branching advantage

Fix integer branching and role bounds $b \geq 2$, $k \geq 0$, $r \geq 0$, and $m \geq 0$, independent of the integer solution depth $D \geq 1$. Measure both procedures in one fixed abstract operation unit, and suppose flat search has branching factor b. Normalize the cost of expanding one flat internal node to one such unit. Counting internal nodes at depths $0, \ldots, D-1$ gives

$$C_{\text{flat}}^{\text{node}}(D) = \sum_{d=0}^{D-1} b^d = \frac{b^D - 1}{b - 1}.$$

For AHR, express each role operation in the same unit. Assume each Scout proposes at most k branches per retained node, the Queen retains at most r, and each retained branch yields at most m Worker successors. Suppose the total Scout, Queen, Worker, and fixed bookkeeping work at one retained node is at most

$$c_{\text{role}} = c_0 + c_U k + c_Q k + c_W r m,$$

where the nonnegative constants c_0, c_U, c_Q, c_W do not depend on D.

Theorem 13.1 (Hierarchical role-work bound)**.** *The AHR role-operation work satisfies*

$$C_{\text{AHR}}^{\text{op}}(D) \leq c_{\text{role}} \sum_{d=0}^{D-1} (rm)^d.$$

If $1 < rm < b$*, then*

$$\frac{C_{\text{AHR}}^{\text{op}}(D)}{C_{\text{flat}}^{\text{node}}(D)} = O\left(\left(\frac{rm}{b}\right)^D\right) \to 0.$$

Proof. At depth 0, at most one node is retained. If at most r Scout branches per retained node survive Queen filtering and each retained branch produces at most m Worker successors, then one retained node has at most rm retained children. Induction on depth therefore gives at most $(rm)^d$ retained nodes at depth d. Expanding each such node costs at most c_{role}

role-operation units, so

$$C^{\mathrm{op}}_{\mathrm{AHR}}(D) \le \sum_{d=0}^{D-1} c_{\mathrm{role}}(rm)^d = c_{\mathrm{role}} \sum_{d=0}^{D-1} (rm)^d.$$

Put $q = rm$. If $1 < q < b$, evaluation of both geometric sums yields

$$\frac{C^{\mathrm{op}}_{\mathrm{AHR}}(D)}{C^{\mathrm{node}}_{\mathrm{flat}}(D)} \le c_{\mathrm{role}} \frac{q^D - 1}{q - 1} \frac{b - 1}{b^D - 1}.$$

For $D \ge 1$,

$$q^D - 1 \le q^D \quad \text{and} \quad b^D - 1 = b^D(1 - b^{-D}) \ge b^D(1 - b^{-1}).$$

Consequently

$$\frac{C^{\mathrm{op}}_{\mathrm{AHR}}(D)}{C^{\mathrm{node}}_{\mathrm{flat}}(D)} \le \frac{c_{\mathrm{role}}(b - 1)}{(q - 1)(1 - b^{-1})} \left(\frac{q}{b}\right)^D.$$

The prefactor is independent of D, proving the stated big-O bound. Because $0 < q/b < 1$, the final power tends to zero. □

If $rm = 0$, no retained node has a retained child. Hence only the root can be expanded and

$$C^{\mathrm{op}}_{\mathrm{AHR}}(D) \le c_{\mathrm{role}}, \qquad \frac{C^{\mathrm{op}}_{\mathrm{AHR}}(D)}{C^{\mathrm{node}}_{\mathrm{flat}}(D)} \le c_{\mathrm{role}} \frac{b - 1}{b^D - 1} \le c_{\mathrm{role}} b\, b^{-D} \to 0.$$

If $rm = 1$, the same calculation gives $C^{\mathrm{op}}_{\mathrm{AHR}}(D) \le c_{\mathrm{role}} D$. Using $b^D - 1 \ge b^D(1 - b^{-1})$ gives

$$\frac{C^{\mathrm{op}}_{\mathrm{AHR}}(D)}{C^{\mathrm{node}}_{\mathrm{flat}}(D)} \le c_{\mathrm{role}} D \frac{b - 1}{b^D - 1} \le c_{\mathrm{role}} b D\, b^{-D}.$$

For $a_D = Db^{-D}$,

$$\frac{a_{D+1}}{a_D} = \frac{D + 1}{bD} = \frac{1 + 1/D}{b} \longrightarrow \frac{1}{b} < 1,$$

so the ratio test gives $Db^{-D} \to 0$. This is a cost result, not a correctness result. Completeness additionally requires that Queen pruning preserve at least one valid path. It is also a conditional asymptotic comparison under the displayed cost model, not a measured wall-clock speedup. In particular, the constants, memory traffic, batching behavior, and the cost of obtaining a smaller effective branching factor must be measured in an implementation. When $rm \ge b$, this result asserts no asymptotic advantage.

13.2 Fixed parameters versus inference cost

Weight sharing can make parameter count independent of NKT. It does not make inference time independent of NKT. A careful claim is:

$$\text{parameter count} = O(1) \text{ in recurrent depth}, \qquad \text{inference work} = O(NKT \cdot c_{\mathrm{step}}).$$

Calling AHR an "$O(1)$ reasoner" without naming the variable is therefore mathematically misleading.

13.3 Detached one-step gradients

Let recurrent state z_t be detached before $z_{t+1} = F_\theta(\text{stopgrad}(z_t), x)$, as in the efficient training motivation inherited from recurrent reasoning and decoupled-gradient methods [18, 29].

Proposition 13.2 (Local autograd memory)**.** *Assume each step has a fixed-size transition module, a local loss is backpropagated before advancing, and that step's graph is then released. The reverse-graph depth and peak activation memory are independent of total unrolled depth T, up to fixed module/state constants and deliberately retained heads.*

Proof. At step t, $\text{stopgrad}(z_t)$ is a leaf with no graph edge to steps $0, \dots, t-1$. The graph created for F_θ and the local loss therefore contains only the fixed computation for step t. Immediate backpropagation traverses this graph, after which releasing it frees its activations before step $t+1$ is constructed. Thus the maximum number of simultaneously retained step graphs is one, so both graph depth and peak activation storage are bounded by constants determined by one transition, one state, and retained heads, not by T. □

The immediate-backward condition is essential. If all T local losses and their step graphs are retained for one final backward call, peak storage can again be $O(T)$ even though recurrent states are detached.

For the next result, let $z_t \in \mathbb{R}^d$ and $\theta \in \mathbb{R}^p$. Under the stated row-gradient convention, the relevant types are

object	shape
$J_t = \partial F_\theta / \partial z$	$d \times d$
$G_t = \partial F_\theta / \partial \theta$	$d \times p$
$\nabla_z \mathcal{L}(z_T)$	$1 \times d$
$\nabla_\theta \mathcal{L}$	$1 \times p$

so every product in the proposition is dimensionally well typed.

Proposition 13.3 (Exact detached-gradient bias bound)**.** *Let $T \geq 1$, let $z_{t+1} = F_\theta(z_t, x)$ for $t = 0, \dots, T-1$, let z_0 be independent of θ, and let the scalar terminal loss be $\mathcal{L}(z_T)$ with no other direct dependence on θ. Assume F_θ and $\mathcal{L}$ are differentiable at every point used by this trajectory, and define*

$$J_t = \frac{\partial F_\theta}{\partial z}(z_t, x), \qquad G_t = \frac{\partial F_\theta}{\partial \theta}(z_t, x).$$

Using row-gradient convention, the full gradient is

$$\nabla_\theta \mathcal{L} = \sum_{j=0}^{T-1} \nabla_z \mathcal{L}(z_T) J_{T-1} J_{T-2} \cdots J_{j+1} G_j,$$

where the product is the identity when $j = T-1$. Detaching at every step and backpropagating only the terminal loss retains only the $j = T-1$ term. If compatible submultiplicative operator/vector norms and constants $0 \leq L < 1$, $G \geq 0$, and $C \geq 0$ satisfy

$$\|J_t\| \leq L, \qquad \|G_t\| \leq G \quad \text{for every } t \in \{0, \dots, T-1\}, \qquad \|\nabla_z \mathcal{L}(z_T)\| \leq C,$$

then the omitted gradient Δ_T obeys

$$\|\Delta_T\| \le CG \sum_{r=1}^{T-1} L^r = CG\,\frac{L(1-L^{T-1})}{1-L} \le CG\,\frac{L}{1-L}.$$

Proof. Let $D_t = \partial z_t/\partial\theta$. Because z_0 is independent of θ, $D_0 = 0$. The chain rule gives

$$D_{t+1} = J_t D_t + G_t.$$

We claim

$$D_T = \sum_{j=0}^{T-1} J_{T-1} J_{T-2} \cdots J_{j+1} G_j.$$

For $T = 1$, $D_1 = J_0 D_0 + G_0 = G_0$, which is the formula with an empty Jacobian product. If it holds at T, then

$$\begin{aligned} D_{T+1} &= J_T D_T + G_T \\ &= J_T \sum_{j=0}^{T-1} J_{T-1} \cdots J_{j+1} G_j + G_T \\ &= \sum_{j=0}^{T} J_T J_{T-1} \cdots J_{j+1} G_j, \end{aligned}$$

where the $j = T$ product is empty. This proves the claim by induction. Multiplying by the terminal row gradient yields the displayed full-gradient formula.

Under stepwise detachment, z_{T-1} is treated as constant in the final transition, so the terminal backward pass yields $\nabla_z \mathcal{L}(z_T) G_{T-1}$. Subtracting this retained term from the full gradient gives

$$\Delta_T = \sum_{j=0}^{T-2} \nabla_z \mathcal{L}(z_T) J_{T-1} \cdots J_{j+1} G_j.$$

The triangle inequality and submultiplicativity imply

$$\begin{aligned} \|\Delta_T\| &\le \sum_{j=0}^{T-2} \|\nabla_z \mathcal{L}(z_T)\| \prod_{k=j+1}^{T-1} \|J_k\|\, \|G_j\| \\ &\le CG \sum_{j=0}^{T-2} L^{T-1-j}. \end{aligned}$$

Setting $r = T - 1 - j$ changes the final sum to $\sum_{r=1}^{T-1} L^r$. Its geometric-sum value and the inequality $1 - L^{T-1} \le 1$ give the claimed bound. □

The bound controls bias when transitions are contractive, but it does not make the bias zero. When L is close to one, the factor $L/(1-L)$ can be large; when $L \ge 1$, this argument provides no depth-uniform guarantee.

Example 13.4 (Two transitions without a skipped chain-rule term). For $T = 2$,

$$D_1 = G_0, \qquad D_2 = J_1 D_1 + G_1 = J_1 G_0 + G_1.$$

Therefore the full terminal-loss gradient is

$$\nabla_\theta \mathcal{L} = \nabla_z \mathcal{L}(z_2) J_1 G_0 + \nabla_z \mathcal{L}(z_2) G_1.$$

Detaching z_1 before the second transition treats z_1 as constant and retains only the second term. The omitted term is exactly $\nabla_z \mathcal{L}(z_2) J_1 G_0$, which is the $T = 2$ instance of Δ_T.

Part III

Complete and Domain-Specific Instantiations

Chapter 14

Finite Modus-Ponens AHR

14.1 Independent semantics

Let

$$\Phi_{\mathrm{MP}} = \{\mathsf{P}, \mathsf{Q}, \mathsf{P} \to \mathsf{Q}\}, \qquad \Gamma_0 = \{\mathsf{P}, \mathsf{P} \to \mathsf{Q}\},$$

and let $x_0 = (\Gamma_0, \mathsf{Q})$. A Boolean valuation is a map $v : \{\mathsf{P}, \mathsf{Q}\} \to \{0, 1\}$, extended to implication by

$$v(\varphi \to \psi) = \max\{1 - v(\varphi), v(\psi)\}.$$

Write $v \models \Gamma_0$ when $v(\gamma) = 1$ for every $\gamma \in \Gamma_0$, and write

$$\Gamma_0 \models \varphi \quad \Longleftrightarrow \quad \forall v,\ v \models \Gamma_0 \Longrightarrow v(\varphi) = 1.$$

The answer space and semantic solution relation are

$$X_{\mathrm{MP}} = \{x_0\}, \qquad Y_{\mathrm{MP}} = \Phi_{\mathrm{MP}},$$

$$\mathrm{Sol}_{\mathrm{MP}} = \{(x_0, \varphi) : \varphi = \mathsf{Q} \text{ and } \Gamma_0 \models \varphi\}.$$

This relation is defined by truth under valuations and contains no proof checker.

Lemma 14.1 (Semantic modus-ponens instance)**.** $\Gamma_0 \models \mathsf{Q}$.

Proof. Let v be arbitrary and assume $v \models \Gamma_0$. Then $v(\mathsf{P}) = 1$ and $v(\mathsf{P} \to \mathsf{Q}) = 1$. By the definition of implication,

$$1 = v(\mathsf{P} \to \mathsf{Q}) = \max\{1 - v(\mathsf{P}), v(\mathsf{Q})\} = \max\{0, v(\mathsf{Q})\} = v(\mathsf{Q}).$$

Thus every valuation satisfying Γ_0 also satisfies Q, which is exactly $\Gamma_0 \models \mathsf{Q}$. □

14.2 Finite certificates and checker

A proof line is a triple (formula, rule, citations). Use the six-line universe

$$\ell_1 = (\mathsf{P}, \mathrm{Asm}, \varnothing), \qquad \ell_2 = (\mathsf{P} \to \mathsf{Q}, \mathrm{Asm}, \varnothing),$$
$$\ell_3 = (\mathsf{Q}, \mathrm{MP}, (1,2)), \qquad \ell_4 = (\mathsf{Q}, \mathrm{Asm}, \varnothing),$$
$$\ell_5 = (\mathsf{Q}, \mathrm{MP}, \varnothing), \qquad \ell_6 = (\mathsf{P}, \mathrm{AC}, (2,3)).$$

Here AC denotes the invalid rule *affirm the consequent*. Let $\mathcal{L}_{\mathrm{MP}} = \{\ell_1, \ldots, \ell_6\}$ and

$$\mathcal{P}_{\mathrm{MP}} = \bigcup_{n=0}^{3} \mathcal{L}_{\mathrm{MP}}^{n}.$$

For $p = (p_1, \ldots, p_n) \in \mathcal{P}_{\mathrm{MP}}$ and $i \in \{1, \ldots, n\}$, write $\mathrm{form}(p_i)$ for the formula in line i. Define $\mathrm{LineOK}_{\Gamma_0}(i, p) = 1$ recursively if and only if exactly one of the following clauses holds:

1. $p_i = (\varphi, \mathrm{Asm}, \varnothing)$ and $\varphi \in \Gamma_0$;
2. $p_i = (\varphi, \mathrm{MP}, (j,k))$, where $1 \le j, k < i$, $\mathrm{LineOK}_{\Gamma_0}(j, p) = \mathrm{LineOK}_{\Gamma_0}(k, p) = 1$, and for some $\psi \in \Phi_{\mathrm{MP}}$,
$$\mathrm{form}(p_j) = \psi, \qquad \mathrm{form}(p_k) = \psi \to \varphi.$$

The recursion is well-founded because every citation index is strictly less than i. For $i \notin \{1, \ldots, n\}$, define $\mathrm{LineOK}_{\Gamma_0}(i, p) = 0$. Define

$$\mathrm{Check}_{\Gamma_0}(p) = 1$$

if and only if $1 \le n \le 3$, every line $i \le n$ is valid, and $\mathrm{form}(p_n) = \mathsf{Q}$. All other inputs receive value zero.

Independently, define $\Gamma_0 \vdash \varphi$ to be the least relation on Φ_{MP} generated by the following two inference rules whenever all displayed formulas belong to Φ_{MP}:

$$\frac{\varphi \in \Gamma_0}{\Gamma_0 \vdash \varphi}\ (\mathrm{Asm}), \qquad \frac{\Gamma_0 \vdash \psi \qquad \Gamma_0 \vdash \psi \to \varphi}{\Gamma_0 \vdash \varphi}\ (\mathrm{MP}).$$

Lemma 14.2 (Checked lines are derivable)**.** *Let $p \in \mathcal{P}_{\mathrm{MP}}$ and $i \in \{1, \ldots, |p|\}$. If*

$$\mathrm{LineOK}_{\Gamma_0}(i, p) = 1,$$

then

$$\Gamma_0 \vdash \mathrm{form}(p_i).$$

Proof. We use strong induction on i. For $i = 1$, the MP clause is impossible because it would require integers satisfying $1 \le j, k < 1$. Therefore a valid first line is an assumption $(\varphi, \mathrm{Asm}, \varnothing)$ with $\varphi \in \Gamma_0$, and the assumption rule gives $\Gamma_0 \vdash \varphi$.

Now fix $i > 1$ and assume the result for every valid line with index less than i. If line i is valid by the assumption clause, the same assumption rule gives the conclusion. Otherwise it is valid by the MP clause. Then there are $j, k < i$ and ψ such that the cited lines are valid,

$$\mathrm{form}(p_j) = \psi, \qquad \mathrm{form}(p_k) = \psi \to \mathrm{form}(p_i).$$

The strong induction hypothesis yields

$$\Gamma_0 \vdash \psi, \qquad \Gamma_0 \vdash \psi \to \mathrm{form}(p_i).$$

Applying the MP inference rule gives $\Gamma_0 \vdash \mathrm{form}(p_i)$. These are the only checker clauses, so the induction is complete. □

Theorem 14.3 (Semantic soundness of the toy calculus)**.** *For every $\varphi \in \Phi_{\mathrm{MP}}$,*

$$\Gamma_0 \vdash \varphi \quad \Longrightarrow \quad \Gamma_0 \models \varphi.$$

Proof. Induct on the finite derivation of $\Gamma_0 \vdash \varphi$.

If the last rule is Asm, then $\varphi \in \Gamma_0$. For any $v \models \Gamma_0$, the definition of satisfaction gives $v(\varphi) = 1$. Therefore $\Gamma_0 \models \varphi$.

If the last rule is MP, there is a formula ψ with derivations of $\Gamma_0 \vdash \psi$ and $\Gamma_0 \vdash \psi \to \varphi$. The induction hypotheses give

$$\Gamma_0 \models \psi, \qquad \Gamma_0 \models \psi \to \varphi.$$

Let $v \models \Gamma_0$ be arbitrary. Then $v(\psi) = 1$ and $v(\psi \to \varphi) = 1$. Expanding implication,

$$1 = v(\psi \to \varphi) = \max\{1 - v(\psi), v(\varphi)\} = \max\{0, v(\varphi)\} = v(\varphi).$$

Hence every model of Γ_0 satisfies φ, so $\Gamma_0 \models \varphi$. □

Corollary 14.4 (Checker soundness)**.** *If* $\mathrm{Check}_{\Gamma_0}(p) = 1$*, then* $\Gamma_0 \vdash \mathsf{Q}$ *and* $\Gamma_0 \models \mathsf{Q}$*.*

Proof. Checker acceptance gives a nonempty $p = (p_1, \ldots, p_n)$ whose final line is valid and has formula Q. The checked-line lemma gives $\Gamma_0 \vdash \mathsf{Q}$, and semantic soundness of the calculus then gives $\Gamma_0 \models \mathsf{Q}$. □

14.3 Exact finite Queen, Worker, and Scout spaces

Use the branch universe

$$\mathcal{K}_{\mathrm{MP}} = \{k_P, k_{P\to Q}, k_{\mathrm{MP}}, k_Q^{\mathrm{asm}}, k_{\mathrm{AC}}, k_{\mathrm{irr}}\},$$

with

$$\mathcal{K}_{\mathrm{rel}} = \{k_P, k_{P\to Q}, k_{\mathrm{MP}}\}, \qquad \mathcal{K}_{\mathrm{bad}} = \mathcal{K}_{\mathrm{MP}} \setminus \mathcal{K}_{\mathrm{rel}}.$$

Define finite coordinate sets

$$\begin{aligned}
\mathcal{B}_Q &= \{0, 1, \ldots, 6\}, & \mathcal{V}_Q &= \{\mathrm{mustVerify}\},\\
\mathcal{D}_Q &= \{\mathrm{undecided}, \mathrm{accept}, \mathrm{reject}, \mathrm{abstain}\}, & \mathcal{G}_{\mathrm{MP}} &= \{\mathrm{derive}(\mathsf{P}), \mathrm{derive}(\mathsf{P}\to\mathsf{Q}), \mathrm{derive}(\mathsf{Q})\},\\
\mathcal{F}_W &= \{\mathrm{none}, \mathrm{localPass}, \mathrm{localFail}\}, & \mathcal{E}_K &= \{0, 1, 2, +\infty\}.
\end{aligned}$$

The role and task-state spaces are exactly

$$\begin{aligned}
Q_{\mathrm{MP}} &= \mathcal{B}_Q \times \mathcal{V}_Q \times \mathcal{D}_Q \times 2^{\mathcal{K}_{\mathrm{MP}}},\\
W_{\mathrm{MP}} &= (\mathcal{G}_{\mathrm{MP}} \cup \{\mathrm{none}\}) \times \mathcal{P}_{\mathrm{MP}} \times \mathcal{F}_W,\\
U_{\mathrm{MP}} &= 2^{\mathcal{K}_{\mathrm{MP}}} \times 2^{\mathcal{K}_{\mathrm{MP}}} \times \mathcal{E}_K^{\mathcal{K}_{\mathrm{MP}}},\\
S_{\mathrm{MP}} &= \{\Gamma_0\} \times \{\mathsf{Q}\} \times \{\mathrm{unchecked}, \mathrm{pass}, \mathrm{fail}\} \times \{\mathrm{running}, \mathrm{stopped}\},\\
H_{\mathrm{MP}} &= Q_{\mathrm{MP}} \times W_{\mathrm{MP}} \times U_{\mathrm{MP}} \times S_{\mathrm{MP}}.
\end{aligned}$$

A Queen state is $q = (b, \nu, d, K_{\mathrm{pruned}})$, a Worker state is $w = (g, p, f)$, and a Scout state is $u = (F_U, B_U, e_U)$.

These are compressed coordinates, not exceptions to the canonical schemas. To display the exact identification, fix singleton symbols $\sigma_0, z_0, r_0, d_0, \pi_0, p_0, n_0, \xi_0$. The corresponding uncompressed spaces are

$$\begin{aligned}
Q_{\mathrm{MP}}^{\mathrm{full}} &= \{\sigma_0\} \times \mathcal{B}_Q \times \mathcal{V}_Q \times \mathcal{D}_Q \times \{z_0\} \times 2^{\mathcal{K}_{\mathrm{MP}}},\\
W_{\mathrm{MP}}^{\mathrm{full}} &= (\mathcal{G}_{\mathrm{MP}} \cup \{\mathrm{none}\}) \times \mathcal{P}_{\mathrm{MP}} \times \{r_0\} \times \{d_0\} \times \mathcal{F}_W \times \{\pi_0\},\\
U_{\mathrm{MP}}^{\mathrm{full}} &= 2^{\mathcal{K}_{\mathrm{MP}}} \times \{\varnothing\} \times 2^{\mathcal{K}_{\mathrm{MP}}} \times \mathcal{E}_K^{\mathcal{K}_{\mathrm{MP}}} \times \{n_0\}^{\mathcal{K}_{\mathrm{MP}}} \times \{\xi_0\}^{\mathcal{K}_{\mathrm{MP}}},\\
S_{\mathrm{MP}}^{\mathrm{full}} &= \{\mathrm{derive}(\mathsf{Q})\} \times \{\Gamma_0\} \times \{\mathrm{unchecked}, \mathrm{pass}, \mathrm{fail}\} \times \{p_0\} \times \{\mathrm{running}, \mathrm{stopped}\}.
\end{aligned}$$

The explicit bijections send

$$\begin{aligned}
(b, \nu, d, K_{\mathrm{pruned}}) &\longmapsto (\sigma_0, b, \nu, d, z_0, K_{\mathrm{pruned}}),\\
(g, p, f) &\longmapsto (g, p, r_0, d_0, f, \pi_0),\\
(F_U, B_U, e_U) &\longmapsto (F_U, \varnothing, B_U, e_U, \boldsymbol{n}_0, \boldsymbol{\xi}_0),\\
(\Gamma_0, \mathsf{Q}, v_{\mathrm{stat}}, r_{\mathrm{stat}}) &\longmapsto (\mathrm{derive}(\mathsf{Q}), \Gamma_0, v_{\mathrm{stat}}, p_0, r_{\mathrm{stat}}),
\end{aligned}$$

where $\boldsymbol{n}_0$ and $\boldsymbol{\xi}_0$ are the constant maps on $\mathcal{K}_{\mathrm{MP}}$. We use the shorter tuples below through these bijections. All projections, transitions, and cardinalities are therefore well typed in the general six-coordinate role and five-coordinate task-state schemas.

Let

$$\mathcal{R}_{\mathrm{MP}} = \{\mathrm{Idle}, \mathrm{Asm}, \mathrm{MP}, \mathrm{AC}\}.$$

The action spaces are

$$\begin{aligned}
A_Q^{\mathrm{MP}} &= (\mathcal{G}_{\mathrm{MP}} \cup \{\mathrm{none}\}) \times 2^{\mathcal{K}_{\mathrm{MP}}} \times \{0,1\} \times \mathcal{D}_Q \times \{0,1\},\\
A_W^{\mathrm{MP}} &= \mathcal{R}_{\mathrm{MP}} \times 2^{\mathcal{K}_{\mathrm{MP}}} \times (\mathcal{L}_{\mathrm{MP}} \cup \{\mathrm{none}\}),\\
A_U^{\mathrm{MP}} &= 2^{\mathcal{K}_{\mathrm{MP}}},\\
A_{\mathrm{MP}} &= A_Q^{\mathrm{MP}} \times A_W^{\mathrm{MP}} \times A_U^{\mathrm{MP}}.
\end{aligned}$$

Write

$$a^Q = (g_A, K_{\mathrm{keep}}, \chi_V, d_A, \chi_S), \quad a^W = (r_A, K_{\mathrm{use}}, \ell_A), \quad a^U = P_U.$$

Proposition 14.5 (Finiteness of the toy AHR)**.** *Every set just defined is finite. In particular,*

$$|\mathcal{P}_{\mathrm{MP}}| = \sum_{n=0}^{3} 6^n = 1 + 6 + 36 + 216 = 259,$$

$$|Q_{\mathrm{MP}}| = 7 \cdot 1 \cdot 4 \cdot 2^6 = 1792, \quad |W_{\mathrm{MP}}| = 4 \cdot 259 \cdot 3 = 3108,$$

and

$$|U_{\mathrm{MP}}| = 2^6 \cdot 2^6 \cdot 4^6.$$

Consequently H_{MP} and A_{MP} are finite.

Proof. The six-line alphabet has 6^n strings of length n, which gives the first calculation. A six-element set has 2^6 subsets and there are 4^6 maps from $\mathcal{K}_{\mathrm{MP}}$ to the four-element set $\mathcal{E}_K$. The displayed cardinalities now follow from the multiplication rule for finite Cartesian products. The remaining coordinate and action sets are finite products or finite unions of explicitly finite sets. □

14.4 Exact compatibility and dynamics

Define line metadata by

ℓ	ℓ_1	ℓ_2	ℓ_3	ℓ_4	ℓ_5	ℓ_6
$\beta(\ell)$	k_P	$k_{P\to Q}$	k_{MP}	k_Q^{asm}	k_{MP}	k_{AC}
$g(\ell)$	derive(P)	derive(P → Q)	derive(Q)	derive(Q)	derive(Q)	derive(P)
$r(\ell)$	Asm	Asm	MP	Asm	MP	AC

For current proof p, define

$$p^+ = \begin{cases} p, & \ell_A = \text{none}, \\ p \parallel \ell_A, & \ell_A \in \mathcal{L}_{\mathrm{MP}} \text{ and } |p| < 3, \end{cases}$$

where $\parallel$ denotes append. Let

$$K_{\mathrm{cand}} = F_U \cup P_U, \qquad K_{\mathrm{drop}} = K_{\mathrm{cand}} \setminus K_{\mathrm{keep}},$$

and define the prospective verifier status

$$v^+ = \begin{cases} \text{pass}, & \chi_V = 1 \text{ and } \mathrm{Check}_{\Gamma_0}(p^+) = 1, \\ \text{fail}, & \chi_V = 1 \text{ and } \mathrm{Check}_{\Gamma_0}(p^+) = 0, \\ \text{unchecked}, & \chi_V = 0 \text{ and } \ell_A \neq \text{none}, \\ v_{\mathrm{stat}}, & \chi_V = 0 \text{ and } \ell_A = \text{none}. \end{cases}$$

The third case invalidates stale verification whenever the certificate changes.

For $h = (q, w, u, s)$ and $a = (a^Q, a^W, a^U)$, declare the tuple incompatible when p^+ is undefined. When p^+ is defined, let $(q, w, u, s, a) \in \mathcal{A}_{\mathrm{QWS}}^{\mathrm{MP}}$ if and only if all of the following finite predicates hold:

1. $K_{\mathrm{keep}} \subseteq K_{\mathrm{cand}} \cap \mathcal{K}_{\mathrm{rel}}$ and $K_{\mathrm{drop}} \subseteq \mathcal{K}_{\mathrm{bad}}$;
2. if $\ell_A = \text{none}$, then $r_A = \mathrm{Idle}$ and $K_{\mathrm{use}} = \varnothing$;
3. if $\ell_A \in \mathcal{L}_{\mathrm{MP}}$, then

$$g_A = g(\ell_A), \quad r_A = r(\ell_A), \quad K_{\mathrm{use}} = \{\beta(\ell_A)\} \subseteq K_{\mathrm{keep}};$$

4. $d_A = \text{accept}$ implies $v^+ = \text{pass}$;
5. $\chi_S = 1$ implies either
$$d_A = \text{accept and } v^+ = \text{pass},$$
or
$$d_A \in \{\text{reject}, \text{abstain}\} \text{ and } (b = 1 \text{ or } K_{\text{keep}} = \varnothing).$$

Clause 3 admits ℓ_5: it uses the authorized MP role, goal, and branch, but its missing citations remain a checker error. Clauses 1 and 3 exclude ℓ_4 and ℓ_6 at the role-compatibility level.

The dynamic transition
$$(x_0, h, a, h') \in T^{\text{MP}}_{\text{dyn}}$$
holds exactly when the current run status is running, $b \geq 1$, p^+ is defined, and the following updates hold:
$$\begin{aligned} q' &= (b-1, \nu, d_A, K_{\text{pruned}} \cup K_{\text{drop}}), \\ w' &= (g_A, p^+, f^+), \\ u' &= (K_{\text{keep}}, B_U \cup P_U, e'_U), \\ s' &= (\Gamma_0, \mathsf{Q}, v^+, r^+), \end{aligned}$$

where
$$f^+ = \begin{cases} \text{none}, & \ell_A = \text{none}, \\ \text{localPass}, & \ell_A \neq \text{none and } \text{LineOK}_{\Gamma_0}(|p^+|, p^+) = 1, \\ \text{localFail}, & \text{otherwise}, \end{cases}$$
$$e'_U(k) = \begin{cases} 0, & k \in P_U, \\ e_U(k), & k \notin P_U, \end{cases} \qquad r^+ = \begin{cases} \text{stopped}, & \chi_S = 1, \\ \text{running}, & \chi_S = 0. \end{cases}$$

The strict transition relation is
$$T^{\text{MP}}_{\text{AHR}} = T^{\text{MP}}_{\text{dyn}} \cap \{(x_0, h, a, h') : (q, w, u, s, a) \in \mathcal{A}^{\text{MP}}_{\text{QWS}}\}.$$

Let $e_0(k) = 2$ for every $k \in \mathcal{K}_{\text{MP}}$. Initialization is the singleton
$$I_{\text{MP}}(x_0) = \{h_0\},$$
$$h_0 = ((6, \text{mustVerify}, \text{undecided}, \varnothing), (\text{none}, (), \text{none}), (\varnothing, \varnothing, e_0), (\Gamma_0, \mathsf{Q}, \text{unchecked}, \text{running})).$$
These equations completely determine the finite toy transition system.

14.5 Extraction, verification, and domain soundness

For
$$h = ((b, \nu, d, K_{\text{pruned}}), (g, p, f), u, (\Gamma_0, \mathsf{Q}, v_{\text{stat}}, r_{\text{stat}})),$$
define
$$R_{\text{MP}}(h) = \begin{cases} \text{form}(p_{|p|}), & r_{\text{stat}} = \text{stopped}, \ d = \text{accept}, \ \text{and } |p| \geq 1, \\ \bot, & \text{otherwise}. \end{cases}$$

For a terminal strict-legal trajectory $\tau = (h_0, a_0, \ldots, h_L) \in \Omega^{\mathrm{MP}}_{\mathrm{AHR}}(x_0)$ whose terminal Worker proof is p_L and whose terminal Queen decision is d_L, define

$$V_{\mathrm{MP}}(x_0, \tau, \varphi) = 1$$

if and only if

$$\begin{aligned} r_{\mathrm{stat},L} &= \text{stopped}, & d_L &= \text{accept}, \quad v_{\mathrm{stat},L} = \text{pass}, \\ R_{\mathrm{MP}}(h_L) &= \varphi, & \mathrm{Check}_{\Gamma_0}(p_L) &= 1. \end{aligned}$$

As in the general construction, define $V^{\mathrm{ext}}_{\mathrm{MP}}$ on raw toy trajectories to equal V_{MP} on $\Omega^{\mathrm{MP}}_{\mathrm{AHR}}(x_0)$ and zero outside that strict domain. For the bounded toy objective, set

$$\mathrm{stepCost}_{\mathrm{MP}}(x_0, h, a, h') = 1 \quad \text{for every } (h, a, h') \in H_{\mathrm{MP}} \times A_{\mathrm{MP}} \times H_{\mathrm{MP}}, \qquad B_{\mathrm{MP}} = 6,$$

and set the three role-specific soft energies to zero. The general hard alignment, transition, verifier, extraction, and budget energies then define the toy's full extended-real objective. In particular, every allowed transition has positive cost and every budget-feasible trajectory has at most six transitions.

Proposition 14.6 (Nonaccepting stops cannot extract an answer)**.** *Let $\tau \in \Omega^{\mathrm{MP}}_{\mathrm{AHR}}(x_0)$. If the terminal Queen decision is* reject *or* abstain*, then $R_{\mathrm{MP}}(h_L) = \bot$ and $V_{\mathrm{MP}}(x_0, \tau, \varphi) = 0$ for every $\varphi \in Y_{\mathrm{MP}}$.*

Proof. Both reject and abstain differ from accept. The acceptance condition in the first branch of R_{MP} therefore fails, so the second branch gives $R_{\mathrm{MP}}(h_L) = \bot$. Because $\bot \notin Y_{\mathrm{MP}}$, one has $R_{\mathrm{MP}}(h_L) \neq \varphi$ for every $\varphi \in Y_{\mathrm{MP}}$. The extraction equality required by $V_{\mathrm{MP}} = 1$ is false; the explicit verifier requirement $d_L = \text{accept}$ is false as well. Hence the verifier returns zero. □

Theorem 14.7 (Toy verifier domain soundness)**.** *For every $\tau \in \Omega^{\mathrm{MP}}_{\mathrm{AHR}}(x_0)$ and every $\varphi \in Y_{\mathrm{MP}}$,*

$$V_{\mathrm{MP}}(x_0, \tau, \varphi) = 1 \quad \Longrightarrow \quad (x_0, \varphi) \in \mathrm{Sol}_{\mathrm{MP}}.$$

Proof. Assume $V_{\mathrm{MP}}(x_0, \tau, \varphi) = 1$. Then the terminal certificate p_L passes the checker. By checker soundness, $\Gamma_0 \models \mathsf{Q}$. Checker acceptance also says that the final formula of p_L is Q. Extraction and verifier acceptance give

$$\varphi = R_{\mathrm{MP}}(h_L) = \mathrm{form}((p_L)_{|p_L|}) = \mathsf{Q}.$$

Thus both defining conditions $\varphi = \mathsf{Q}$ and $\Gamma_0 \models \varphi$ hold. By the independently defined semantic relation, $(x_0, \varphi) \in \mathrm{Sol}_{\mathrm{MP}}$. □

14.6 Valid and invalid trajectories

Let $K_* = \{k_P, k_{P \to Q}, k_{\mathrm{MP}}\}$. The following six actions, with unlisted Worker components determined by the metadata table and Idle used when no line is appended, define a strict

trajectory from h_0:

t	P_U	K_{keep}	g_A	ℓ_A	χV	d_A	χS
0	K_*	K_*	none	none	0	undecided	0
1	$\varnothing$	K_*	derive(P)	none	0	undecided	0
2	$\varnothing$	K_*	derive(P)	ℓ_1	0	undecided	0
3	$\varnothing$	K_*	derive($\mathsf{P} \to \mathsf{Q}$)	ℓ_2	0	undecided	0
4	$\varnothing$	K_*	derive(Q)	ℓ_3	0	undecided	0
5	$\varnothing$	K_*	none	none	1	accept	1

The terminal certificate is $p_* = (\ell_1, \ell_2, \ell_3)$, its checker status is pass, the terminal budget is zero, and

$$R_{\text{MP}}(h_6) = \mathsf{Q}, \qquad V_{\text{MP}}(x_0, \tau_*, \mathsf{Q}) = 1.$$

Replacing ℓ_3 at time 4 by ℓ_5 preserves role-level compatibility: the Queen authorizes the Q goal, the Worker declares MP, and it uses k_{MP}. Nevertheless,

$$\text{LineOK}_{\Gamma_0}(3, (\ell_1, \ell_2, \ell_5)) = 0$$

because ℓ_5 has no earlier citation pair. Hence the prospective status at verification is fail and an accepting stop is QWS-incompatible. The resulting raw attempt is outside $\Omega^{\text{MP}}_{\text{AHR}}(x_0)$, so the rejection extension $V^{\text{ext}}_{\text{MP}}$ equals zero. This exact counterexample shows that alignment among the Queen, Worker, and Scout and proof validity are distinct.

Chapter 15

Finite Constraint Satisfaction and Temporal Reasoning

15.1 CSP substrate

A finite CSP is

$$\mathcal{P}_{\mathrm{CSP}} = (\mathcal{X}, \{D_i\}_{i=1}^{n}, \mathcal{C}, Q_{\mathrm{out}}),$$

where $\mathcal{X} = \{x_1, \ldots, x_n\}$, every domain D_i is finite, $\mathcal{C} = \{c_1, \ldots, c_m\}$, and Q_{out} specifies which part of a satisfying assignment is returned. Each constraint c_j has scope $S_j \subseteq \mathcal{X}$ and predicate

$$R_j : \prod_{x_i \in S_j} D_i \to \{0, 1\}.$$

The indexed product denotes assignments to the named variables in S_j (formally, functions choosing one value in each corresponding domain), so the predicate does not depend on an arbitrary ordering of the set S_j. For assignment a, define hard energy

$$E_{\mathrm{CSP}}(a) = \sum_{j=1}^{m} \mathbf{1}\{R_j(a|_{S_j}) = 0\}.$$

Proposition 15.1 (Exact CSP energy)**.** *For every complete assignment $a \in \prod_{i=1}^{n} D_i$,*

$$E_{\mathrm{CSP}}(a) = 0 \quad \Longleftrightarrow \quad R_j(a|_{S_j}) = 1 \text{ for every } j \in \{1, \ldots, m\}.$$

Proof. Each indicator $\mathbf{1}\{R_j(a|_{S_j}) = 0\}$ is either zero or one and is therefore nonnegative. If every constraint predicate equals one, every indicator is zero, so their sum is zero.

Conversely, suppose the sum is zero. If some predicate R_k were zero, its indicator would equal one. Since all other indicators are nonnegative, this would imply

$$E_{\mathrm{CSP}}(a) = \sum_{j=1}^{m} \mathbf{1}\{R_j(a|_{S_j}) = 0\} \geq \mathbf{1}\{R_k(a|_{S_k}) = 0\} = 1,$$

contradicting $E_{\mathrm{CSP}}(a) = 0$. Hence every predicate equals one. □

Thus zero energy is equivalent to exact-checker acceptance. For a CSP-solving task whose

contract accepts any satisfying assignment, verifier validity, rather than equality with one stored reference assignment, is the correct success criterion when solutions are non-unique [3, 5].

15.2 Role mapping

Table 15.1: AHR roles in a finite CSP.

Role	Mathematical responsibility
Scout	Propose variable-value choices, factor messages, decompositions, or repair moves.
Worker	Construct or revise a partial assignment and maintain local consistency information.
Queen	Choose subproblems, allocate radius/budget, prune by admissible bounds, invoke the checker, accept, or abstain.
Verifier	Evaluate all hard predicates independently of the learned score.

15.3 Temporal and revised constraints

Fix a represented finite horizon $t \in \{0, \dots, T\}$. Temporal CSPs use variables $x_{t,i} \in D_i$ and legal transition predicates

$$(x_t, x_{t+1}) \in \mathcal{T}_t.$$

A rule activated at time $\tau_r \in \{0, \dots, T\}$ is represented by

$$\forall t \in \{\tau_r, \dots, T\}, \qquad r_t(x_t) = 1.$$

Thus mid-trajectory rule revision is not an exception to AHR; it changes the constraint coordinate c_t and therefore the strict transition relation from that time onward.

Chapter 16

Proof-Assistant Search

Let s_t be a formal proof state and $\mathcal{A}(s_t) = \{a_1, \ldots, a_k\}$ a finite generated action set. A proof assistant provides local checking

$$V_{\text{local}}(s_t, a) = 1$$

when a typechecks. Only such a locally accepted pair lies in the domain of the successor function T. Local acceptance does not guarantee proof closure. Define the exact completion set without applying T outside its domain:

$$\mathcal{C}(s_t, a) = \begin{cases} \{\tau : \tau \text{ is a legal continuation from } T(s_t, a) \text{ to a closed proof}\}, & V_{\text{local}}(s_t, a) = 1, \\ \varnothing, & V_{\text{local}}(s_t, a) = 0. \end{cases}$$

With $\inf \varnothing := +\infty$, let $\text{cost}_{\text{PA}}(\tau) \in \mathbb{R}_{\geq 0}$ be the proof-assistant cost of continuation τ and define exact branch energy

$$E^*_{\text{branch}}(s_t, a) = \inf_{\tau \in \mathcal{C}(s_t, a)} \text{cost}_{\text{PA}}(\tau).$$

A finite-valued learned or heuristic energy $\widehat{E}_{\text{branch}}(s_t, a) \in \mathbb{R}$ estimates this quantity:

$$\widehat{E}_{\text{branch}}(s_t, a) \approx E^*_{\text{branch}}(s_t, a).$$

If at least one action has a proof-closing continuation, the ideal local choice is

$$a^* \in \arg\min_{a \in \mathcal{A}(s_t)} E^*_{\text{branch}}(s_t, a).$$

For a nonempty generated action set, if a ranker induces $p(a \mid s) \propto e^{-\widehat{E}(s,a)}$, its entropy is

$$H(A \mid s) = -\sum_{a \in \mathcal{A}(s)} p(a \mid s) \log p(a \mid s).$$

Low entropy is useful only when it concentrates on locally accepted actions that lead to closure. Lean-AHR is this instantiation: Scouts generate tactics or lemmas, Workers construct proof-state transitions, the Queen controls top-k expansion and budget, and Lean is the proof authority [4].

The main theory obligations are candidate-set coverage, safe pruning, domain soundness of

the checker, and a useful approximation to continuation cost. High top-k completion on a templated family is empirical evidence, not a completeness theorem for general theorem proving.

Chapter 17

Symbolic and Diophantine Proof Search

A symbolic proof-search instantiation fixes an alphabet Σ, grammar $\mathcal{G}$, proof calculus $\mathcal{R}$, and deterministic verifier. A partial token sequence is a Worker construction; grammar-compatible continuations are Scout branches; the Queen controls normalization, case splitting, bounds, and beam retention.

For prefix p, let $\mathcal{N}(p)$ be its grammar-valid next tokens and define $V_{\text{prefix}}(p) = 1$ exactly when no irreparable scope, syntax, or arithmetic error is detected. Thus zero denotes prefix rejection. The strict frontier is

$$\{p\sigma : \sigma \in \mathcal{N}(p),\ V_{\text{prefix}}(p\sigma) = 1\}.$$

Sound prefix rejection gives safe pruning only when every rejected prefix truly has no valid proof completion. The next proposition states the bounded conclusion precisely.

Proposition 17.1 (Bounded symbolic completeness)**.** *Assume Σ is finite, $L < \infty$, and a target valid proof $p^* \in \Sigma^{\leq L}$ exists, where $\Sigma^{\leq L}$ denotes all strings over Σ of length at most L. Suppose:*

1. *every grammar-valid next token at every retained prefix is eventually enumerated;*
2. *prefix pruning rejects only prefixes having no valid proof completion;*
3. *the Worker realizes each enumerated token append; and*
4. *the final verifier accepts every proof that is valid in the fixed calculus.*

Then bounded AHR search eventually enumerates and accepts p^*.

Proof. Write $p^* = \sigma_1 \cdots \sigma_n$ with $n \leq L$, and let $p_i = \sigma_1 \cdots \sigma_i$ for $i = 0, \ldots, n$, where p_0 is empty. We prove by induction that every p_i is eventually reached and retained. The empty prefix p_0 is the initial search node. Assume p_i is reached for some $i < n$. Because p_i extends to the valid proof p^*, assumption 2 forbids pruning it. The next token σ_{i+1} is grammar-valid along p^*, so assumption 1 eventually enumerates it. Assumption 3 realizes the append and reaches p_{i+1}. The same sound-pruning argument retains that prefix because it still extends to p^*.

Finite induction reaches $p_n = p^*$. Assumption 4 then makes the final verifier accept it. Finiteness of $\Sigma^{\leq L}$ guarantees that the bounded exhaustive schedule is over a finite candidate universe; the coverage assumptions identify the target within it. □

There is no algorithm that always terminates and correctly decides, for every multivariate polynomial with integer coefficients, whether the equation $P(z_1, \ldots, z_n) = 0$ has an integer solution. This is the negative solution to Hilbert's tenth problem; the usual natural-number and integer-solution forms are computably equivalent [30]. This does not rule out semidecision completeness for positive instances: an unbounded enumerator may eventually find every existing integer tuple and its certificate while running forever on some no-solution inputs. The bounded proposition above avoids both claims by fixing a finite proof universe.

The Diophantine Slot Machine and AHR-CSP arithmetic reports instantiate this certificate-first view: learned or heuristic search proposes proof tokens, congruence cases, valuations, or factor assignments, while exact arithmetic checks certificates [6, 7, 8]. Domain-specific number-theory claims remain separate from the general AHR theorems.

Chapter 18

Sudoku and Structured Prediction

For Sudoku, let the clue grid be $x \in \{0, 1, \ldots, 9\}^{81}$, where zero marks an empty cell. Let V be the 81 cells, let $D = \{1, \ldots, 9\}$, and let E be the set of unordered pairs $\{u, v\}$ of distinct cells sharing a row, column, or 3×3 box. Thus each peer conflict is counted once. A candidate $y \in D^{81}$ has conflict energy

$$E_{\text{Sudoku}}(y; x) = \sum_{\{u,v\} \in E} \mathbf{1}\{y_u = y_v\} + \sum_{v : x_v \neq 0} \mathbf{1}\{y_v \neq x_v\}.$$

Proposition 18.1 (Exact Sudoku energy)**.** *For $y \in D^{81}$,*

$$E_{\text{Sudoku}}(y; x) = 0$$

if and only if y agrees with every nonzero clue in x and no two peer cells have the same digit.

Proof. Every summand in both sums is a nonnegative 0–1 indicator. If y agrees with every clue and all peer digits differ, every displayed predicate inside an indicator is false, so every summand is zero and the energy is zero.

Conversely, suppose the energy is zero. If some clue cell v had $y_v \neq x_v$, the corresponding clue indicator would be one and the total energy would be at least one, a contradiction. Hence all clues agree. If some peer pair $\{u, v\} \in E$ had $y_u = y_v$, its conflict indicator would likewise be one, again forcing total energy at least one. Hence all peer digits differ. These are precisely the stated validity conditions because y is already a complete assignment in D^{81}. More explicitly, each row contains nine pairwise distinct values drawn from the nine-element set D; therefore each digit occurs exactly once in that row. The identical counting argument applies to every column and every 3×3 box. Thus the peer condition is exactly the usual completed-Sudoku condition. □

A pheromone tensor $\varrho \in \mathbb{R}^{81 \times 9}$ biases cell-digit choices. Scouts perform fast stochastic local exploration, Workers consolidate peer-graph evidence, and the Queen changes slower strategy, halting, and multi-colony selection. Exact Sudoku verification remains independent of the belief field.

This instantiation supplied the original dynamical picture of AHR [2]. The general mathematical theory is broader: ϱ need not be a neural tensor, and AHR need not be limited to structured prediction.

Chapter 19

Empirical Evidence and Its Exact Scope

19.1 Evidence-audit rule

This chapter records what saved local and repository artifacts demonstrate. It does not promote measurements into axioms. Each claim names the evaluated split, the relevant metric definition, and the principal limitation. The artifacts were audited at repository snapshot `0e57d184a118` for the consolidated verified-reasoning archive and `722d820c1e09` for the original `ahr` repository [9, 10]. The local hybrid notebook is byte-identical to the latter snapshot. The local pure-neural notebook is a distinct file version, so its result is anchored to its local SHA-256 digest and saved history rather than silently attributed to the committed source text. Exact local paths, version comparisons, and full SHA-256 digests for every private source used in this chapter are recorded in the accompanying provenance manifest [12].

Three distinctions are essential:

1. a neural prediction, an exact repaired answer, and an answer that matches one stored reference are different outcomes;
2. a validation measurement used for checkpoint selection is not an independent test measurement; and
3. an implementation called "AHR-lite" need not instantiate the full formal Queen, Worker, and Scout state and compatibility relation.

19.2 Sudoku evidence

19.2.1 Early archive

The early local Sudoku archive contains 24 dated run directories from January and February 2026, with configurations and several checkpoints. The surviving metadata show a recurrent Worker, a slower Queen, stochastic Scout proposals, deep supervision, and a Sudoku constraint penalty. One checkpoint records the field `best_solved_long=0.05625`, but the archive does not preserve a sufficiently precise metric contract, split manifest, and end-to-end evaluation

report to interpret that number as a publication-quality solve rate. It is therefore evidence of implementation history, not a quantitative result used by this theory.

19.2.2 AHR-lite hybrid notebook

The saved hybrid notebook used a 147,042-parameter recurrent model with a Worker update at every step, a Queen update every four steps, a pheromone-like logit state, and an annealed Gaussian perturbation called the Scout [10, 11]. The Gaussian perturbation is not an explicit persistent Scout state with its own typed policy and compatibility clauses. This is therefore accurately described as an AHR-lite or HRM-style precursor, not the complete formal architecture.

The recorded run did not use the public Sudoku-extreme data. It fell back to synthetically generated puzzles: 5,000 training, 500 validation, and 500 test examples. The notebook generated the three lists separately but applied no explicit duplicate or cross-split-overlap check. Puzzle uniqueness was also not checked. On the 500-example saved test evaluation, the notebook reports:

metric	value
composite notebook loss	1.233612
cell reference-match rate	0.788355
whole-board reference-match rate	0.075072
last-four-step stability rate	0.207854
stable and reference-exact rate	0.073336
mean Sudoku conflict energy	9.884160

The saved loss combines deep-supervised cross-entropy, an additional blank-cell cross-entropy, a peer regularizer, an entropy term, and a last-step stability term; it is not a standalone negative log-likelihood. Whole-board reference match means that all 81 predicted digits equal the stored target. Last-four-step stability means that the discrete predictions at the final four recurrent steps are identical. Because the synthetic puzzles were not proved unique, a valid completion can disagree with the stored target.

The same notebook saved a 250-puzzle stress table:

metric	raw neural	hybrid
whole-board reference match	0.076	0.108
notebook validity flag	0.180	1.000
mean conflict energy	9.936	0.000
mean wall time per puzzle (seconds)	0.057375	1.065677

The hybrid column is not a pure-neural solve rate. For each of 16 stochastic colonies, the notebook invoked an exact minimum-remaining-values backtracking solver, using neural logits only to order candidate digits. The saved "exact-repair" field equals one on all 250 puzzles because backtracking found a completion in every case. The field name `solved_exactly` in the notebook means that exact backtracking succeeded; it does not mean equality with the stored target.

The notebook's standalone validity flag checks rows, columns, and boxes. Clues are clamped by the neural model and are never overwritten by the backtracking solver; moreover, the

reported hybrid conflict energy is zero and that energy contains a clue-disagreement term. Thus the 100% hybrid validity result is evidence that the exact repair path produced valid completions for this 250-puzzle synthetic sample. It is not evidence that the neural system solved 100% of the puzzles unaided.

19.2.3 Pure-neural Sudoku-extreme run

A later saved local notebook used the public `sapientinc/sudoku-extreme` dataset, whose card reports about 3.83 million training and 423 thousand test puzzles, exact deduplication, unique solutions, and mathematical inequivalence between train and test puzzles [16, 11]. The run used 1,000,000 examples from the augmented official-training corpus for training and 50,000 different augmented-row indices from that same corpus for validation. The executed local source first generated ten Sudoku-symmetry variants per official training puzzle and only then randomly split augmented rows. The two index sets are disjoint, but the split is not grouped by original puzzle; it therefore does not prevent different symmetry variants of one source puzzle from appearing on both sides.

The pure-neural notebook source at pinned GitHub commit `722d820c1e09` contains a replacement group-preserving split function. However, its saved output payload is identical to the local row-split notebook's output payload and lacks the diagnostic printed by that replacement function. Thus the archive does not evidence a completed run of the grouped version. The reported result is conservatively interpreted according to the executed local row-split source. Its validation set is consequently neither an official split nor documented as source-puzzle-independent. The model had 708,450 parameters and 40 recurrent steps.

The complete local 100-epoch history records its best *validation-selected* point at epoch 93:

validation metric	value
loss	1.225052
cell reference-match rate	0.837523
zero-conflict-energy rate	0.474780
pure exact rate	0.474780
mean conflict energy	8.898020

The pure-exact predicate requires both all 81 digits to match the target and zero clue/peer conflict energy. The official test loader contained 422,786 examples, but no completed independent test evaluation was saved. Therefore 47.478% is an augmented-row validation result used in model selection, not a source-puzzle-independent validation estimate or a test-set result. The model also uses a slower Queen-like update, a faster Worker-like update, pheromone logits, and Gaussian Scout noise rather than the complete formal Scout object.

19.3 Triple-role AHR-CSP evidence

The Phase 5r9 AHR-CSP artifact is the clearest saved experiment with explicit three-scale recurrence: two Queen cycles, two Worker cycles, and four Scout steps, with evaluation noise disabled [13]. Its synthetic satisfiable data comprise 4,096 training, 1,024 validation, and 1,024 test instances drawn from the same generator families: all-different, Boolean SAT, and bounded arithmetic.

On the 1,024-example test split, the saved report gives

metric	value
assignment-coordinate reference accuracy	0.924729
whole-assignment reference match	0.711914
vectorized hard-valid rate	1.000000
mean vectorized hard energy	0.000000

The vectorized hard-valid computation covers all 1,024 test outputs. A separate CPU verifier checked 64 test outputs and accepted all 64; the reported CPU-valid rate of 1.0 therefore has denominator 64, not 1,024. The gap between 71.1914% reference match and 100% hard validity is expected for CSPs with multiple satisfying assignments and directly motivates the distinction between semantic validity and equality with one reference label.

This experiment supports feasibility of the Queen/Worker/Scout recurrent decomposition and exact constraint measurement on these bounded synthetic families. It does not establish the completeness assumptions, safe-pruning premise, asymptotic work bound, or generalization to arbitrary CSPs.

19.4 Abstaining multi-family ranker

The saved 4,998,161-parameter multi-family ranker has 17 output labels and split sizes 1,440 training, 360 validation, and 360 test [14]. On both validation and test, it records 340 correct answers, zero wrong answers, and 20 abstentions:

$$\text{coverage} = \frac{340}{360} = 0.944444\ldots, \qquad \text{hard-valid rate among answers} = 1.$$

The equality follows by direct division: $340/360 = 17/18$. These results are consistent with the formal abstention contract on the saved synthetic task bundle. They do not estimate performance on open-domain reasoning, and the artifact explicitly forbids clinical use.

19.5 Lean proof-search evidence, including the negative result

The Lean-AHR v3.6 artifact used 50 ProofNet test rows with indices 50–99. ProofNet is a 371-example benchmark of undergraduate mathematics statements, informal proofs, and Lean 3 formal statements [17, 15]. The parse/retrieval audit translated and retrieved 33 of 50 rows. The subsequent full bridge made 769 Lean calls and produced exact Lean-verified target proofs for 2 of 50 rows. The saved failure counts are 29 verification failures, 10 unknown identifiers, 5 type mismatches, 2 typeclass-synthesis failures, and 2 unsolved-goal failures; together with the 2 successes they sum to all 50 rows:

$$29 + 10 + 5 + 2 + 2 + 2 = 50.$$

The batch began as a held-out offset, but its translation failures were mined and repaired before the v3.6 rerun. The final 2/50 rate must consequently be treated as a development-set-style rerun, not an untouched test estimate. This negative result is important: exact local checking and a useful search architecture do not by themselves solve proof translation

or proof-script synthesis.

19.6 What the evidence establishes

The saved artifacts establish only the following empirical facts:

- Queen-like slow recurrence, Worker-like fast construction, and pheromone-like memory have been trained in Sudoku precursors;
- an explicit Queen/Worker/Scout recurrent configuration has achieved zero measured hard energy on one 1,024-instance synthetic AHR-CSP test split;
- exact repair can convert neural Sudoku guidance into valid solutions on a small synthetic sample, at additional compute cost;
- hard validity can differ substantially from reference equality on non-unique tasks;
- abstention can eliminate measured errors on a bounded synthetic ranker split by reducing coverage; and
- broad Lean proof synthesis remains largely unsolved in the audited artifact.

No saved experiment yet proves that learned Queen pruning is safe, that Scout coverage satisfies the probability lower bounds of the completeness theorem, that the full formal compatibility relation is enforced during neural training, or that AHR has an empirical asymptotic advantage over matched HRM, EBM, and flat-search baselines. Those remain explicit research obligations.

Part IV

Learning Theory and Research Program

Chapter 20

Learning Hierarchical Energies

20.1 Parameterized energy

Let $E_\theta = E_{Q,\theta} + E_{W,\theta} + E_{U,\theta} + E_{\text{soft},\theta}$ while hard verification, extraction, budget, and strict transition feasibility remain external. For verified and invalid candidate pairs z^+, z^- and a fixed margin $\gamma > 0$, a margin loss is

$$\mathcal{L}_{\text{margin}}(\theta) = \mathbb{E}\big[\max\{0, \gamma + E_\theta(z^+) - E_\theta(z^-)\}\big].$$

A role-aware objective may add

$$\mathcal{L} = \lambda_Q \mathcal{L}_Q + \lambda_W \mathcal{L}_W + \lambda_U \mathcal{L}_U + \lambda_A \mathcal{L}_{\text{align}} + \lambda_M \mathcal{L}_{\text{margin}} + \lambda_C \mathbb{E}[\text{cost}_x(\tau)].$$

The canonical value $+\infty$ for hard verifier failure is not differentiated. Training may use verifier outcomes to label candidates or may introduce a separately named finite surrogate loss, but that surrogate does not replace the hard verification condition in the emission rule.

20.2 Ranking consistency target

For a distribution over states, assume each $\mathcal{A}(s)$ is finite and nonempty. Let $E^*(s,a) \in \mathbb{R} \cup \{+\infty\}$ be verified completion cost, assume $\min_{a \in \mathcal{A}(s)} E^*(s,a) < +\infty$, and let $\widehat{E}(s,a) \in \mathbb{R}$ be a finite learned score. The useful statistical target is not merely action-label accuracy. It is low regret relative to verified completion cost:

$$\mathcal{R}(\widehat{E}) = \mathbb{E}\left[E^*(s, \widehat{a}(s)) - \min_{a \in \mathcal{A}(s)} E^*(s,a)\right], \qquad \widehat{a}(s) = \text{Tie}\left(\arg\min_{a \in \mathcal{A}(s)} \widehat{E}(s,a)\right).$$

Because the optimal completion cost is finite, the difference is well-defined in $[0, +\infty]$; choosing an action with no verified completion gives infinite regret rather than the undefined expression $+\infty - (+\infty)$. Search-level evaluation should measure completion, verifier calls, compute, and abstention at fixed correctness. A learned energy that ranks local actions accurately but increases total proof-search calls has failed the intended objective.

20.3 Calibration and abstention

Let $p_{\mathrm{succ}}(x, B)$ estimate the probability that bounded search will produce a verified solution. A calibrated stopping rule may abstain when

$$p_{\mathrm{succ}}(x, B) < \theta$$

or when the expected value of another search cycle is below its compute cost. Calibration must be evaluated against actual verifier outcomes and distinguish epistemic uncertainty, search exhaustion, and verifier incompleteness.

Chapter 21

Open Mathematical Problems

Open Problem 21.1 (Repairable alignment). Characterize metrics d_{align} for which every sufficiently low-distance state-action tuple can be repaired into $\mathcal{A}_{\text{QWS}}$ with bounded cost while preserving a valid solution path.

Open Problem 21.2 (Infinite trajectory spaces). Choose natural topologies, such as prefix, proof-tree, edit, or graph-path topologies, under which realistic verified candidate sets are compact and energies are lower semicontinuous, or under which proper energies have finite sublevel sets.

Open Problem 21.3 (Stochastic convergence). Prove rates, not only asymptotic completeness, for energy-guided QWS search under admissible Queen bounds and nonuniform Scout coverage.

Open Problem 21.4 (Learnability). Derive capacity or sample-complexity bounds for hierarchical trajectory energies that preserve verifier-aligned margins under distribution shift.

Open Problem 21.5 (Incomplete verifiers). Quantify the tradeoff between sound but incomplete verification and abstention frequency. A true answer may exist while no certifiable trajectory exists in the representation.

Open Problem 21.6 (Global usefulness). Construct an energy that penalizes locally valid but globally useless Worker moves without pruning every completion of an optimal proof.

Open Problem 21.7 (Factorized inference). Determine when the decomposition $E_Q + E_W + E_U + E_{\text{align}}$ supports exact dynamic programming, admissible lower bounds, or controlled approximation error.

Open Problem 21.8 (Translator soundness). For natural-language input, formalize a typed translator whose uncertainty and ambiguity are exposed to the Queen and whose malformed outputs force abstention rather than silent reinterpretation.

Open Problem 21.9 (Pheromone fixed points). Under stochastic deposits and bounded clamping, characterize invariant distributions, metastable false attractors, and conditions under which exploration prevents premature concentration.

Chapter 22

Falsifiable AHR Research Program

22.1 What the theory already establishes

Under explicit finite or compactness assumptions, the document establishes existence of a verified minimizer; the finite base theory additionally gives a deterministic selection. It also establishes soundness transfer, safe abstention, compatibility coherence, safe lower-bound pruning, conditional coverage completeness, perturbation robustness, and bounded pheromone memory.

22.2 What must be tested

The following claims are empirical and should be tested against flat, random, heuristic, and non-hierarchical energy baselines:

- QWS factorization reduces verifier calls at fixed verified completion.
- Pheromone memory improves long-range search rather than merely storing shortcut labels.
- Three time scales outperform matched two-level and flat recurrence under equal parameters and compute.
- Learned Queen lower bounds remain safe or expose quantified failure rates.
- Scout novelty improves coverage without destroying sample efficiency.
- Abstention is calibrated under representation and verifier shift.

22.3 Minimal ablation lattice

Every serious AHR experiment should compare:

$$\text{flat search} \longrightarrow \text{energy ranking} \longrightarrow \text{Queen/Worker} \longrightarrow \text{Queen/Worker/Scout}$$

$$\text{without pheromone} \longrightarrow \text{with pheromone}$$

$$\text{without hard verifier} \longrightarrow \text{with hard verifier and abstention.}$$

The last comparison is not a license to deploy an unverified system; it is an analysis of where correctness originates.

Chapter 23

The Unified Definition

In the finite base theory, including its positive step-cost and finite-budget assumptions, for $\mathcal{C}_B(x) \neq \varnothing$, let $\mathcal{M}_B(x)$ denote the minimizers of $E_{\mathrm{AHR}}(x, \cdot, \cdot)$ on $\mathcal{C}_B(x)$, and select $(\tau^*, y^*) = \mathrm{Tie}(\mathcal{M}_B(x))$.

$$\mathrm{AHR}_B(x) = \begin{cases} y^*, & \mathcal{C}_B(x) \neq \varnothing, \\ \perp, & \mathcal{C}_B(x) = \varnothing, \end{cases}$$

where

$$\mathcal{C}_B(x) = \{(\tau, y) : \tau \in \Omega_{\mathrm{AHR}}(x),\ R(h_L) = y,\ V(x, \tau, y) = 1,\ \mathrm{cost}_x(\tau) \leq B\},$$

$$T_{\mathrm{AHR}} = T_{\mathrm{dyn}} \cap \{(x, (q, w, u, s), a, h') : (q, w, u, s, a) \in \mathcal{A}_{\mathrm{QWS}}\},$$

and

$$E_{\mathrm{AHR}} = E_Q + E_W + E_U + E_{\mathrm{align}} + E_T + E_V + E_R + E_B.$$

In one sentence:

AHR is a bounded, verifier-constrained, Queen/Worker/Scout-compatible energy search over structured reasoning trajectories.

The mathematical identity of AHR lies in the simultaneous presence of:

1. primitive Queen, Worker, and Scout role spaces;
2. the compatibility relation $\mathcal{A}_{\mathrm{QWS}}$;
3. hierarchical multi-timescale dynamics;
4. energy-based ranking with hard feasibility;
5. independent verification and abstention;
6. optional stigmergic shared memory linking exploration to later action.

Appendix A

Notation

Table A.1: Canonical AHR notation.

Symbol	Meaning
X, Y	Problem and answer spaces.
$\mathrm{Sol} \subseteq X \times Y$	External semantic solution relation.
$G, F, M, \mathcal{P}, C$	Goals, facts, memory, partial constructions, and constraints.
$S = G \times F \times M \times \mathcal{P} \times C$	Structured task state.
Q, W, U	Queen, Worker, and Scout state spaces.
$H = Q \times W \times U \times S$	Full hierarchical state space.
A_Q, A_W, A_U	Role action spaces.
$A = A_Q \times A_W \times A_U$	Joint action space.
$\mathcal{K}$	Scout branch universe.
$\mathcal{A}_{\mathrm{QWS}}$	QWS compatibility relation.
T_{dyn}	Syntactically possible state evolution.
T_{AHR}	Dynamic and QWS-compatible strict transitions.
$I(x)$	Admissible initial states for x.
$\Omega_{\mathrm{raw}}(x), \Omega_{\mathrm{dyn}}(x), \Omega_{\mathrm{AHR}}(x)$	Raw, dynamic, and strict trajectory spaces.
R	Terminal answer/certificate extraction.
V	Independent verifier.
V^{ext}	Total rejection extension of V to raw trajectories.
E_Q, E_W, E_U	Role-specific soft energies.
$E_{\mathrm{align}}, E_T, E_V, E_R, E_B$	Compatibility, transition, verifier, extraction, and budget energies.
$\mathrm{stepCost}, \mathrm{cost}_x$	Problem-dependent step and trajectory costs.
ε	Positive lower bound on the cost of every strict transition in the finite base theory.
B	Hard search budget.
$\Delta(D)$	Probability simplex on a finite nonempty set D.
$\Pi = (\Pi_Q, \Pi_W, \Pi_U)$	Role-factored search policy.
ϱ_t	Stigmergic or pheromone memory, when used.
$\perp$	Abstention/no certified answer.

Appendix B

Hard and Soft Energy Catalogue

Table B.1: Energy semantics.

Term	Hard requirement	Typical finite preference
E_Q	Forbidden Queen decisions are excluded by compatibility or transitions.	Budget allocation, delayed verification, weak arbitration, unnecessary cycles.
E_W	Malformed Worker steps are excluded by compatibility or transitions.	Local detours, progress regression, repair cost, proof length.
E_U	Retained irrelevant branches violate compatibility.	Novelty, uncertainty, duplication, unsupported proposal cost.
E_{align}	Its local summand is 0 on $\mathcal{A}_{\text{QWS}}$ and $+\infty$ outside; the trajectory term is zero exactly when every step is compatible.	Distance to compatibility for learning/repair only.
E_T	Its local summand is 0 on T_{dyn} and $+\infty$ outside; the trajectory term is zero exactly when every transition is dynamic-legal.	None in the canonical hard term; finite smoothness belongs in a separate soft term.
E_V	0 for extended-verifier pass and $+\infty$ for fail or an out-of-domain trajectory.	None in the base emission rule.
E_R	0 when the trajectory claims y, otherwise $+\infty$.	None.
E_B	0 within budget and $+\infty$ beyond it.	Optional finite compute regularizer below the hard cap.

Appendix C

Dependency Map of the Theory

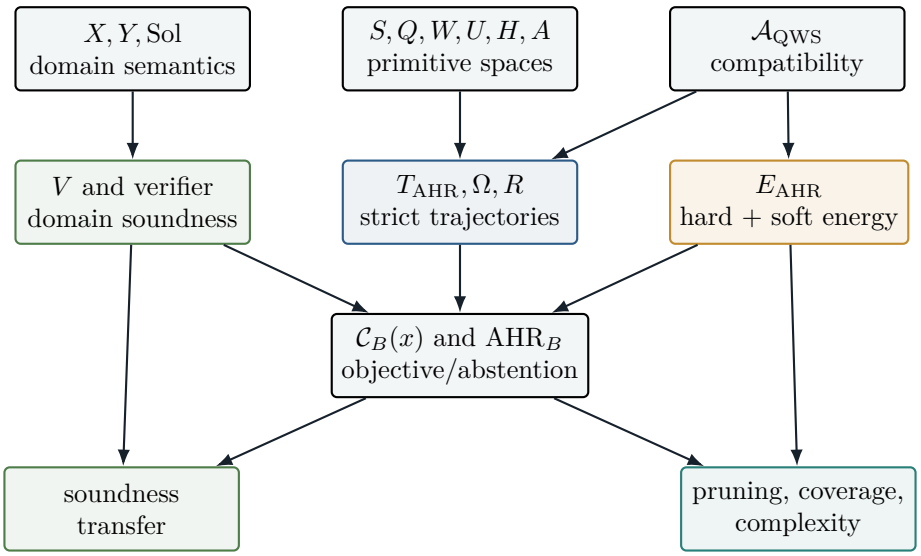


Figure C.1: Logical dependencies. The verifier bridges internal certificates to external semantics; search theorems depend on the objective and on extra coverage/pruning assumptions.

Appendix D

Source Synthesis Map

This monograph integrates the theory corpus by concept rather than concatenating duplicate drafts.

Table D.1: How the AHR paper corpus contributes to the unified theory.

Source family	Integrated contribution	Treatment here
Original AntHill Reasoner paper [2]	Three time scales, pheromone memory, variational prior, stability, detached training, adaptive halting.	Origin and Design Commitments; Primitive Structure; Stigmergic Memory; Complexity and Training; Learning Hierarchical Energies.
Mathematical foundation draft [1]	Semantic relation, exact role spaces, QWS compatibility, energy decomposition, abstention, theorems, finite proof model.	The full foundational and guarantee parts, with domain, notation, and proof corrections.
AHR verified-reasoning reports [3]	Finite typed candidates, independent verifier, repair, branch cost, careful scope claims.	Semantic Contract; Objective and Search; Guarantees; domain instantiations; research program.
AHR-CSP and medical-logic reports [5]	Finite CSP energy, non-unique validity, temporal constraints, rule revision, abstention need.	Finite Constraint Satisfaction and Temporal Reasoning; empirical metrics are not axioms.
Lean-AHR reports [4]	Proof-state search, local versus future validity, top-k branching, entropy view.	Proof-Assistant Search and the evidence ledger.
Diophantine Slot Machine and Brocard AHR papers	Grammar-constrained symbolic search, exact certificates, bounded completeness, number-theory case states.	Symbolic and Diophantine Proof Search; domain claims remain separate.
Saved experiment artifacts [9, 11, 12]	Exact saved metrics, split sizes, architecture fields, verifier counts, failure counts, paths, and SHA-256 digests.	Empirical Evidence and Its Exact Scope, with explicit limitations.

Source family	Integrated contribution	Treatment here
HRM and recurrent-reasoning literature	Multi-timescale recurrent computation and efficient depth.	Origin; Primitive Structure; Complexity and Training. AHR adds a third role, compatibility, stigmergy, and hard verification.
Energy-based learning literature	Extended-real constraints, structured energies, margin learning, inference by minimization.	Stigmergic Memory; Energy Functional; margins; Learning Hierarchical Energies.
ACO/stigmergy literature	Evaporation-deposition memory and distributed search.	Stigmergic Memory and Energy-Based Dynamics, translated into bounded differentiable dynamics.

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